

Selective inference: **methods and applications**

ICSA Applied Statistics Symposium: Short Courses

Zhimei Ren

University of Pennsylvania, Department of Statistics and Data Science

Agenda

1:30 - 2:00 pm **Lecture 1: global null testing**

2:00 - 2:30 pm **Lecture 2: Family-wise error rate and simultaneous coverage**

2:30 - 3:15 pm **Lecture 3: False discovery rate, Benjamini-Hochberg procedure and knockoffs**

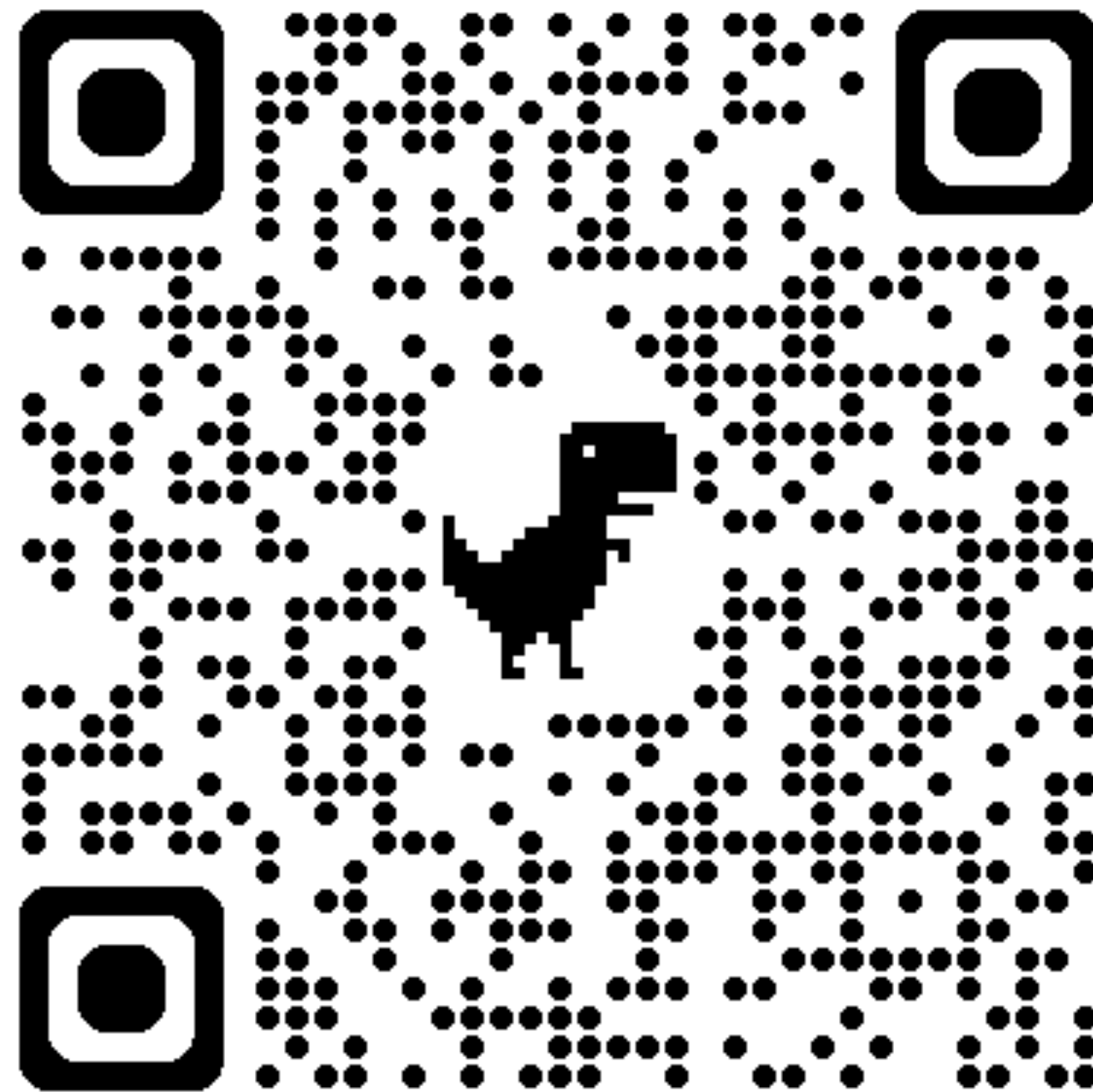
3:15 - 3:30 pm **Checkpoint and break**

3:30 - 4:35 pm **Lecture 4: Adaptive/interactive inference**

4:35 - 5:15 pm **Lecture 5: Post-selection inference**

5:15 - 5:30 pm **Q&A**

Code demo



https://zhimeir.github.io/icsa_codelab/

Lecture 1: Global null testing

Air pollution and disease risks¹

1. Zhou, Hanwen, et al. "Air pollution and risk of 32 health conditions: outcome-wide analyses in a population-based prospective cohort in Southwest China." *BMC medicine* (2024).

Air pollution and disease risks¹

Exposure: air pollution

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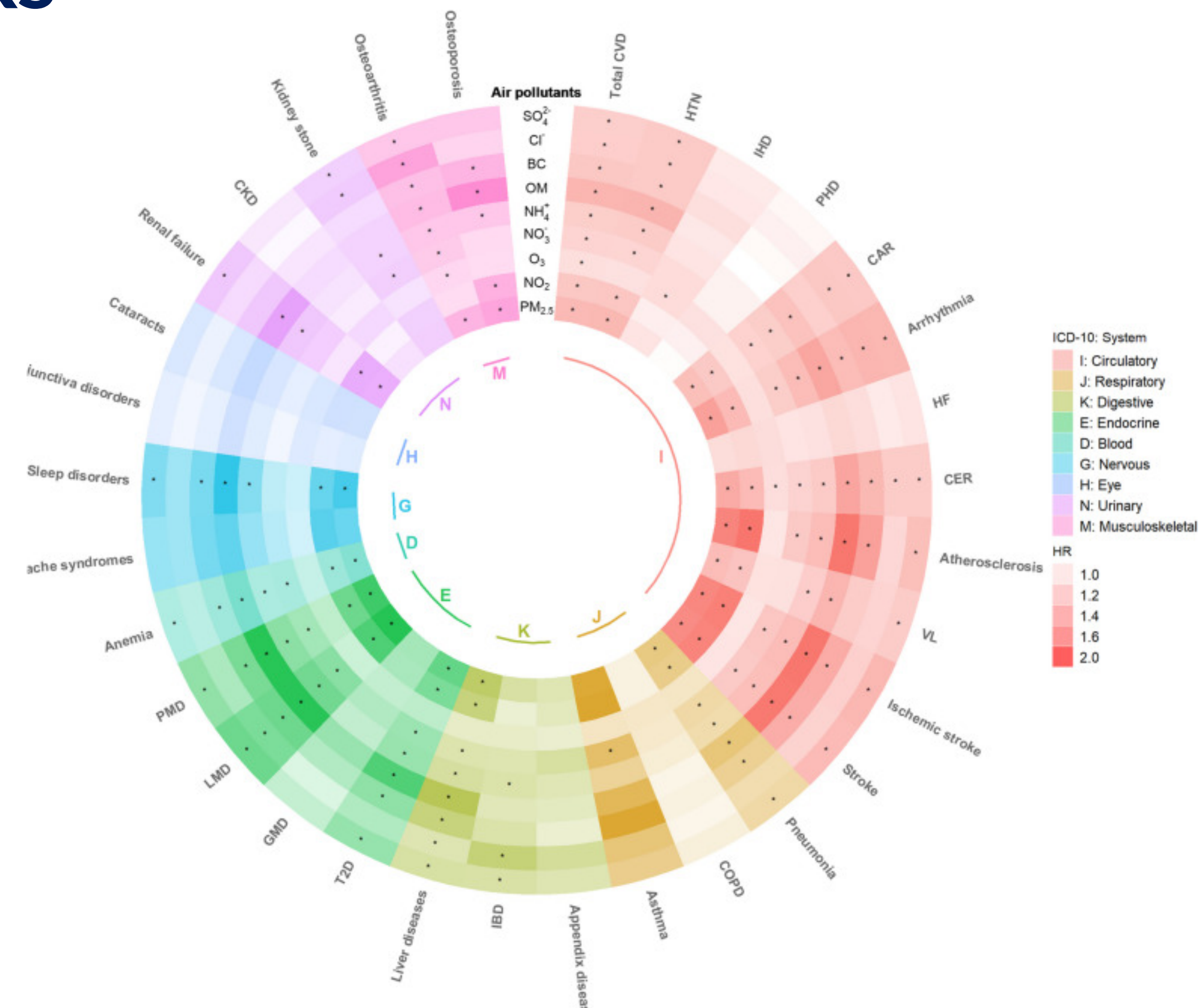
Outcome: 32 health risk factors

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Exposure: air pollution

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Dealing with multiple hypotheses

Asking many questions simultaneously

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For each health risk $j \leadsto$ whether air pollution increases risk j

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For each health risk j $\rightsquigarrow$ whether air pollution increases risk j

$\rightsquigarrow H_j^{(0)}$: air pollution does not increase risk j

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32 such hypotheses

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m Hypotheses of interest: $H_1^{(0)}, H_2^{(0)}, \dots, H_m^{(0)}$

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$$\begin{aligned} \mathbb{P}(\text{making at least 1 rejs}) &= 1 - \mathbb{P}(\text{no rej}) = 1 - (0.95)^m \\ &= 0.401 \text{ when } m = 10 \\ &= 0.642 \text{ when } m = 20 \\ &= 0.806 \text{ when } m = 32 \end{aligned}$$

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A confidence interval for the selected outcomes $\leadsto$ post-selection inference

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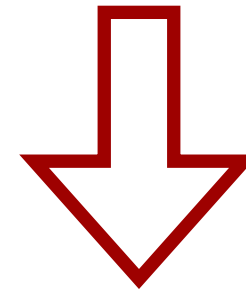
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Global null testing

m hypotheses of interest: $H_1^{(0)}, H_2^{(0)}, \dots, H_m^{(0)}$

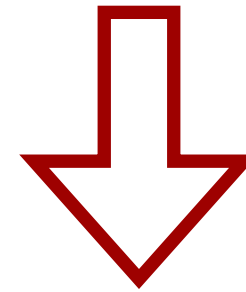
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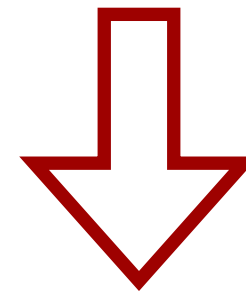
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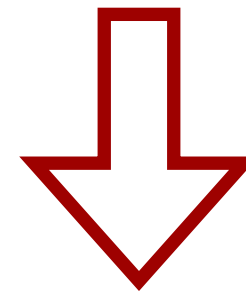


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When $H_{\text{global}}^{(0)}$ is true, want to reject it with probability ≤ 0.05

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Global null hypothesis: $H_{\text{global}}^{(0)} = \cap_{j=1}^m H_j^{(0)}$

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Need to adjust for the multiplicity!

The Bonferroni method

- ▶ Hypotheses: $H_1, \dots, H_m$
- ▶ Global null: $H_{\text{global}} = \cap_{j=1}^m H_j$
- ▶ P-values: $p_1, \dots, p_m$

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reject H_{global} if $\min_j p_j \leq 0.05/m$

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- ▶ **Guarantee:** if H_{global} is true, then $\mathbb{P}(\text{reject } H_{\text{global}}) \leq 0.05$
- ▶ **No assumption on the dependence among the p-values**
- ▶ Power depends on the smallest p-value (strongest evidence)

Fisher's combination method

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$$\text{reject } H_{\text{global}} \text{ if } -2 \sum_{j=1}^m \log(p_j) \geq \chi_{2m,0.95}^2$$

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0.95 quantile of χ_{2m}^2

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0.95 quantile of χ_{2m}^2

- ▶ $-2 \sum_j \log(p_j) \sim \chi_{2m}^2$ under H_{global} when p_j 's are independent
- ▶ More likely to reject w/ many small p-values

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- ▶ Guarantee: if H_{global} is true, then $\mathbb{P}(\text{reject } H_{\text{global}}) \leq 0.05$
- ▶ Requires the p-values to be independent! (Sensitive to violation)
- ▶ Power depends on cumulative strength (e.g., many small p-values)

Cauchy's combination

- ▶ Hypotheses: $H_1, \dots, H_m$
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- ▶ Define $T = \sum_{j=1}^m w_j \tan[(1/2 - p_j)\pi]$, $w_j \geq 0$ & $\sum_j w_j = 1$
- ▶ reject H_{global} if $T \geq C_{0.95}$

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0.95 quantile of the Cauchy dist.

- ▶ $T \sim$ Cauchy distribution under H_{global} when p_j 's are independent
- ▶ More likely to reject w/ many small p-values
- ▶ If $W \sim$ Cauchy and under mild dependence² $\lim_{t \rightarrow \infty} \frac{\mathbb{P}(T > t)}{\mathbb{P}(W > t)} = 1$

2. Liu, Yaowu, and Jun Xie. "Cauchy combination test: a powerful test with analytic p-value calculation under arbitrary dependency structures." *JASA* (2020).

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- ▶ Requires the p-values to be independent, but less sensitive to violation
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Comparison

Method	Assumption	Powerful regime
Bonferroni	Arbitrary dependence	Sparse strong signals
Fisher combination	Independence [sensitive to violation]	Dense weak signals
Cauchy combination	Independence [robust to violation]	Sparse strong signals

Special case of global null testing: evidence combination

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 - A random procedure produces multiple p-values with different random seeds

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- ▶ Consider one hypothesis H
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 - Different test statistics $\leadsto$ different p-values

P-values may be dependent!

Lecture 2: family-wise error rate and simultaneous coverage

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Types of questions being asked:

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Multiple testing procedure

Hypotheses: $H_1, H_2, \dots, H_m$

Multiple testing procedure

CVD

CKD

Sleep disorders

Hypotheses: $H_1, H_2, \dots, H_m$

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Binary decisions: $\delta_1, \delta_2, \dots, \delta_m$

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Binary decisions: $\delta_1, \delta_2, \dots, \delta_m$

Control type-I errors!

Outcome of a multiple testing procedure

- Rejection/selection set $\mathcal{R} \subseteq \{1,2,\dots,m\}$

	Number of false discoveries H_0 true	H_0 false	Number of discoveries Total
Reject	V	S	R
Accept	U	T	
Total	m_0		m

Family-wise error rate

The probability of making **at least one false discoveries** $\mathbb{P}(V \geq 1)$

- ▶ A multiple testing procedure controls the FWER at level 0.05 if $\mathbb{P}(V \geq 1) \leq 0.05$
- ▶ What about rejecting H_j if $p_j \leq 0.05$?

Under independence

Family-wise error rate

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- ▶ P-values: $p_1, \dots, p_m$

Reject H_j if $p_j \leq 0.05/m$

The Bonferroni method

- ▶ Hypotheses: $H_1, \dots, H_m$
- ▶ P-values: $p_1, \dots, p_m$

Reject H_j if $p_j \leq 0.05/m$



Multiplicity correction

The Bonferroni method

- ▶ Hypotheses: $H_1, \dots, H_m$
- ▶ P-values: $p_1, \dots, p_m$

Reject H_j if $p_j \leq 0.05/m$



Multiplicity correction

- ▶ Bonferroni rejection set: $\{j \in \{1, \dots, m\} : p_j \leq 0.05/m\}$
- ▶ Controls FWER at level 0.05
- ▶ No assumptions on the dependence between p-values

A closer look at the Bonferroni method

Why FWER control?

$$\mathbb{P}(V \geq 1) \leq \mathbb{P}(\exists \text{ null } H_j \text{ rejected})$$

A closer look at the Bonferroni method

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Improvement over Bonferroni?

A closer look at the Bonferroni method

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Improvement over Bonferroni?

- ▶ Use knowledge of m_0 ?
- ▶ Under independence

Holm's procedure

Thought experiment

Observe p-values: 0.35, 0.03, 0.001, 0, 0, 0

Bonferroni threshold: $0.05/6$

Holm's procedure

Thought experiment

Observe p-values: 0.35, 0.03, 0.001, 0, 0, 0 

Bonferroni threshold: $0.05/6$

Pretty sure they are non nulls...
Adjust only for 3?

Holm's procedure

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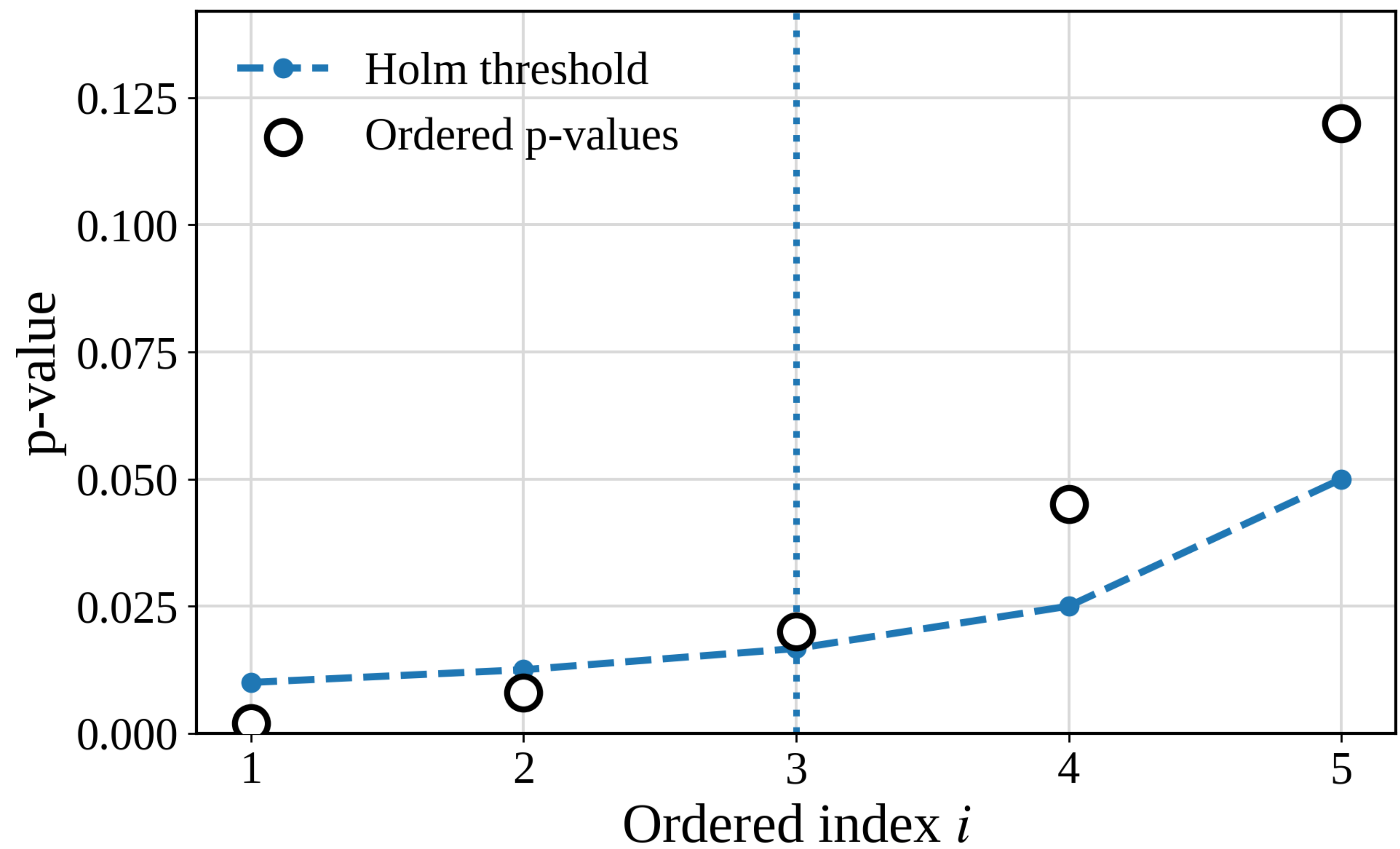
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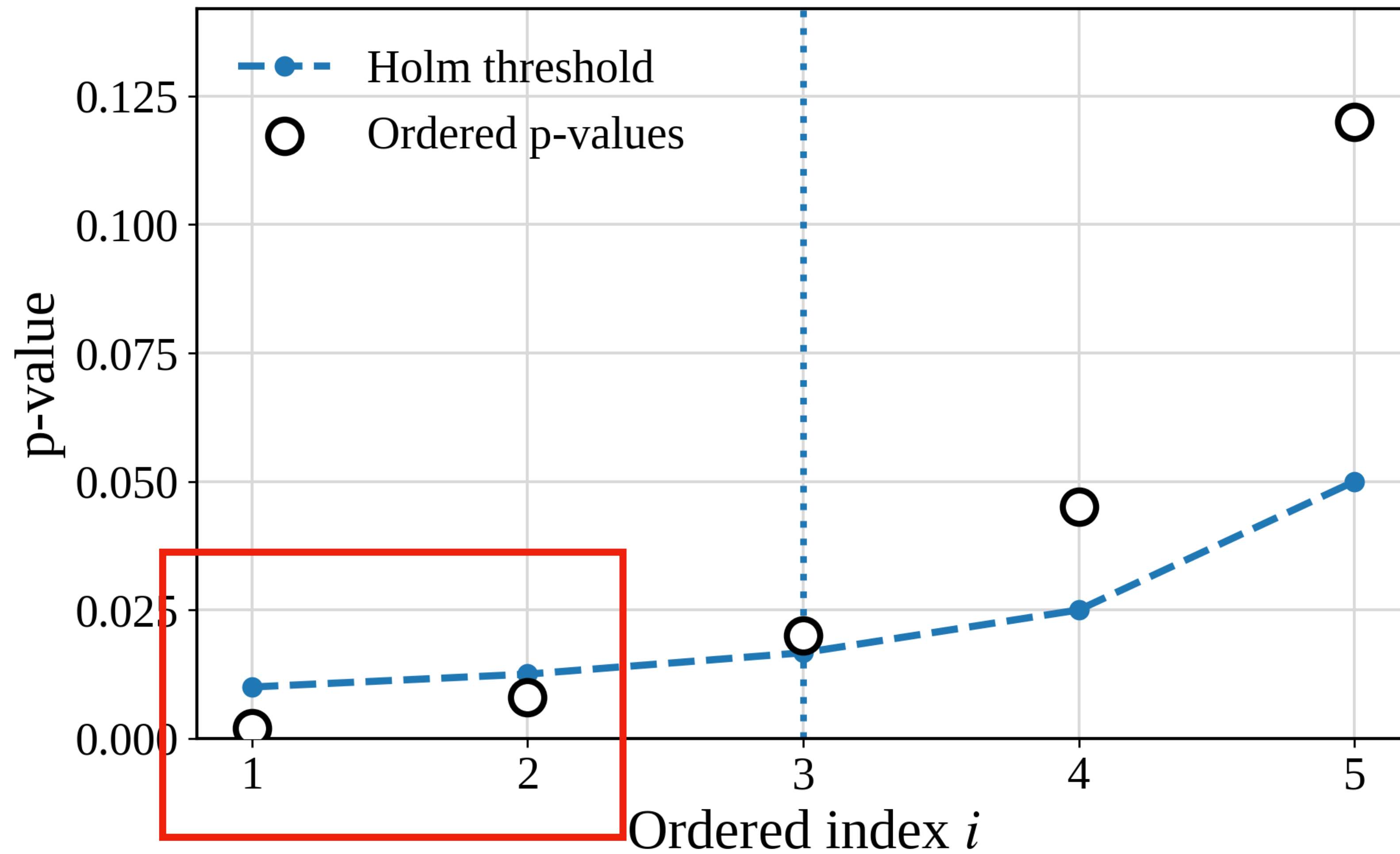
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- ▶ ...
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$$i^* = \min \left\{ j : p_{(j)} > \frac{0.05}{m - j + 1} \right\} \quad \mathcal{R} = \{j : p_j < p_{(j^*)}\}$$

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- ▶ No assumption on the dependence structure among the p-values
- ▶ Makes strictly more rejections than Bonferroni

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Why FWER control?

Let I_0 denote the indices of the nulls. If $\min_{j \in I_0} p_j > \frac{0.05}{I_0}$ then **make no false discoveries**

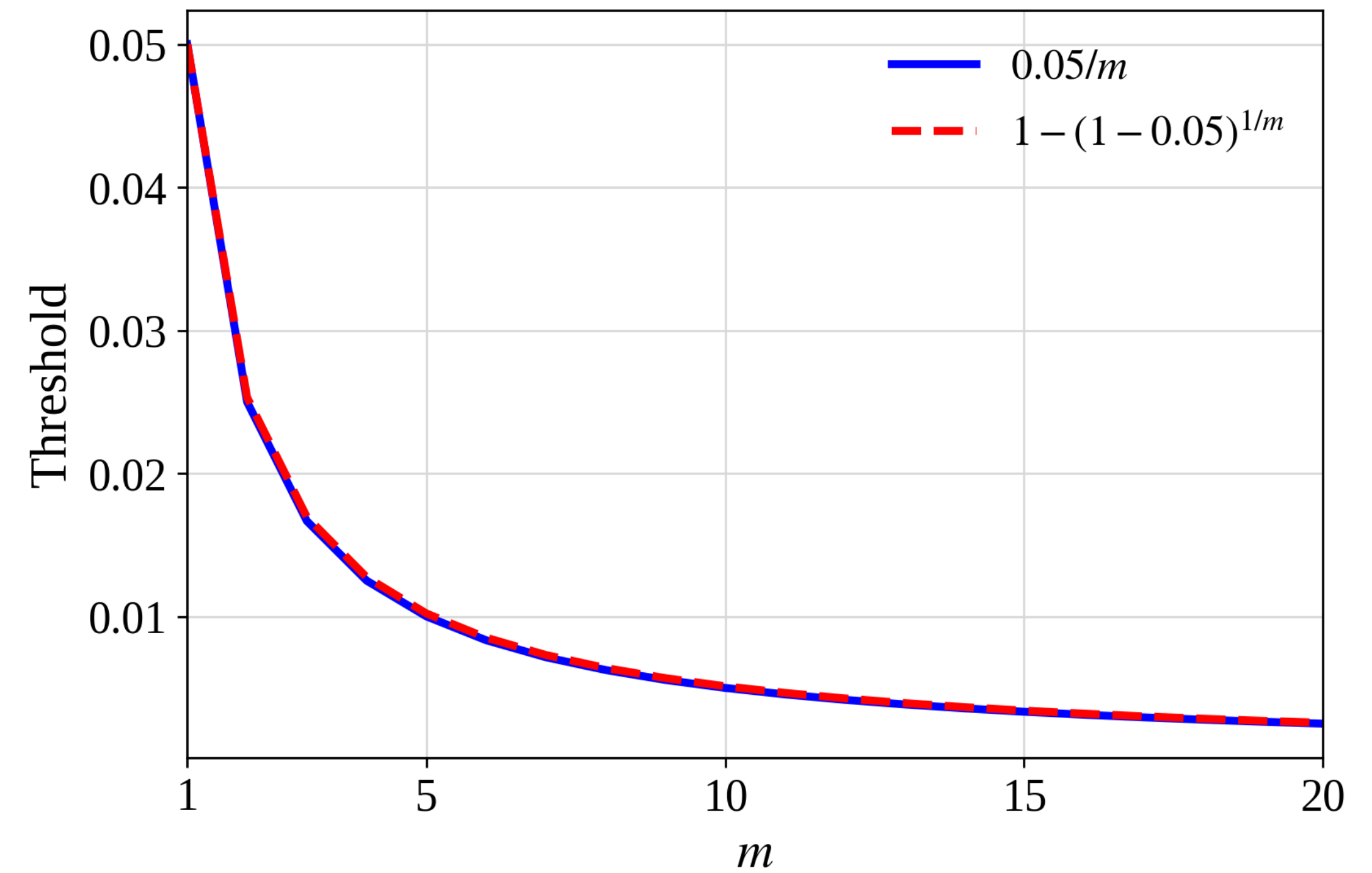
$$\mathbb{P}(V \geq 1) \leq \mathbb{P}\left(\min_{j \in I_0} p_j \geq 0.05/I_0\right) \leq \sum_{j \in I_0} \mathbb{P}(p_j \leq 0.05/I_0) \leq 0.05$$

The Šidák procedure

- ▶ Assume the p-values are independent
- ▶ Reject H_j if $p_j \leq 1 - (1 - \alpha)^{1/m}$

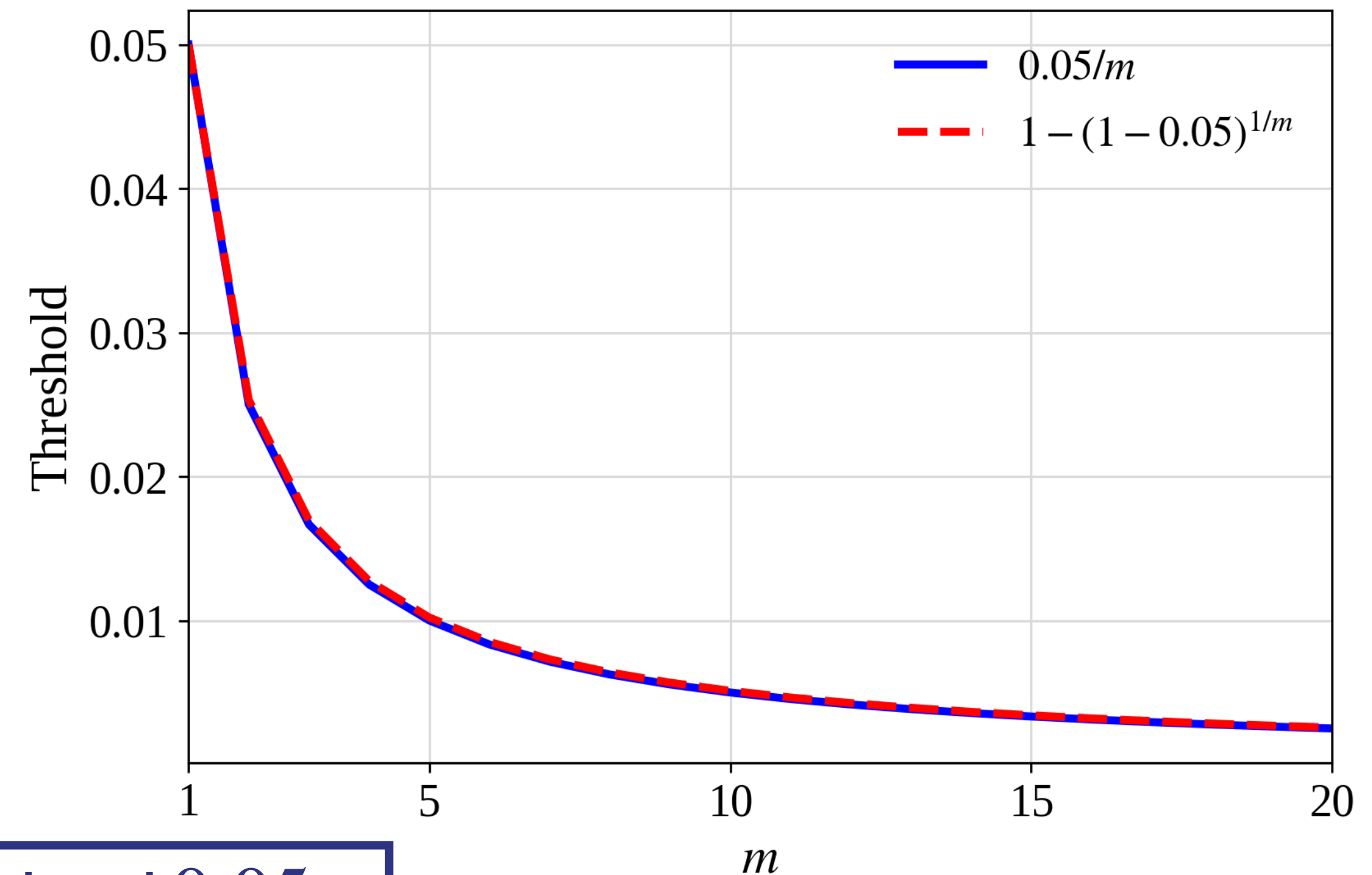
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- ▶ The Šidák procedure controls the FWER at level 0.05
- ▶ Assume the p-values to be independent!
- ▶ Slightly less conservative than Bonferroni

Simultaneous coverage

- Instead of testing for existence of effect
- what if interested in the **range of effect** for each outcome?

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Simultaneous coverage

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$$CI_j = [L_j, U_j]$$

- ▶ For outcome j , use θ_j to denote the effect size (or other quantity of interest)
- ▶ Use the data $\leadsto$ a confidence interval $CI_j = [L_j, U_j]$
- ▶ 95 % **simultaneous coverage**: $\mathbb{P}(\forall j, \theta_j \in [L_j, U_j]) \geq 95 \%$

Simultaneous coverage

Setting

- ▶ For each j , suppose a $(1 - \alpha)$ -marginal confidence interval, CI_j^α satisfying

$$\mathbb{P}(\theta_j \in CI_j^\alpha) \geq 1 - \alpha$$

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No assumption on independence

Lecture 3: FDR, the BH procedure, and knockoffs

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	Number of false discoveries H_0 true	H_0 false	Total Number of discoveries
Reject	V	S	R
Accept	U	T	
Total	m_0		m

False discovery rate (FDR)³

$$\text{FDR} = \mathbb{E}[\text{FDP}] \qquad \text{FDP} = \frac{V}{\max\{R, 1\}}$$

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$$\frac{V}{\max\{R, 1\}} \leq 1 \{V \geq 1\}$$

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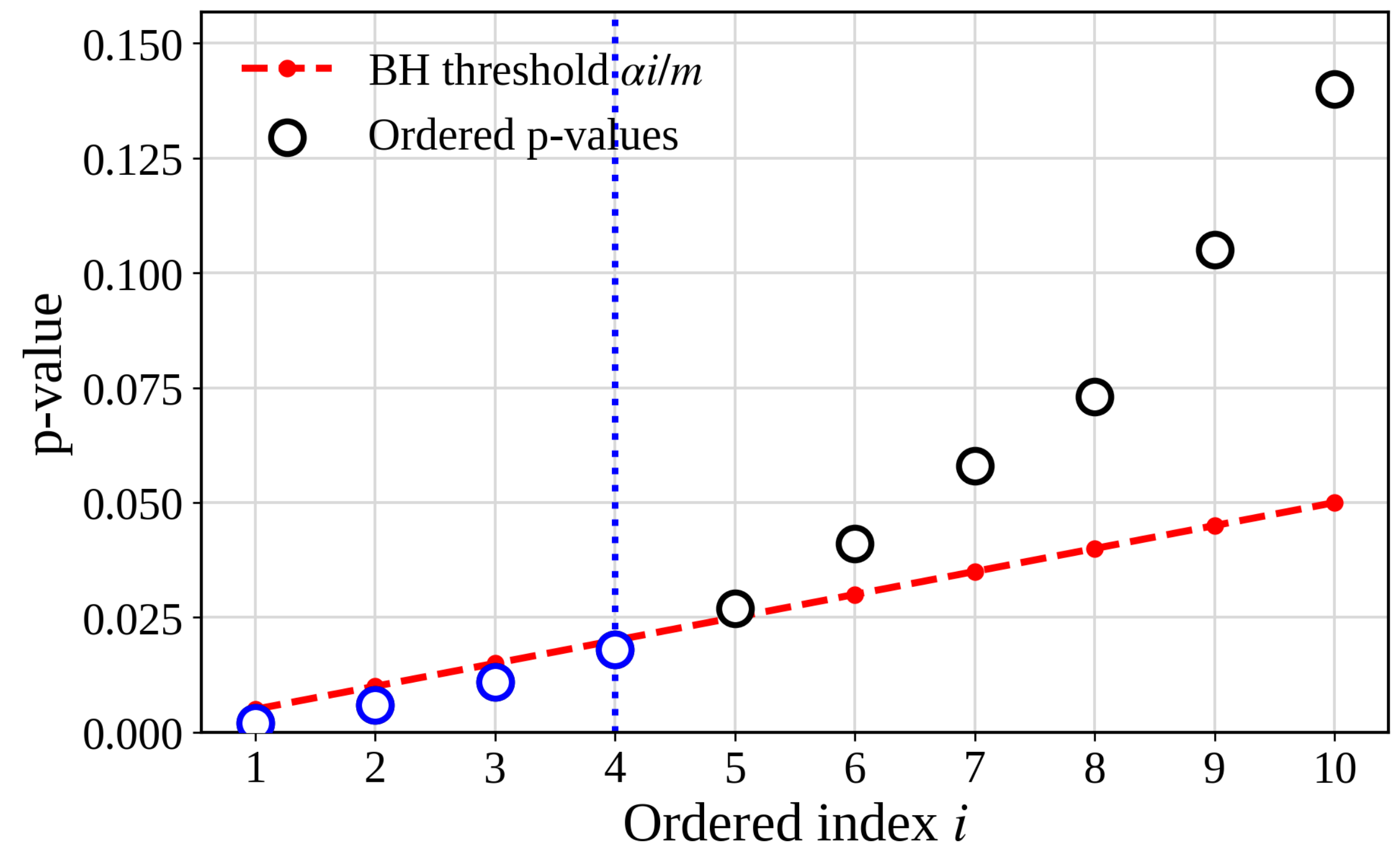
The Benjamini-Hochberg (BH) procedure³

- ▶ Hypotheses: $H_1, H_2, \dots, H_m$
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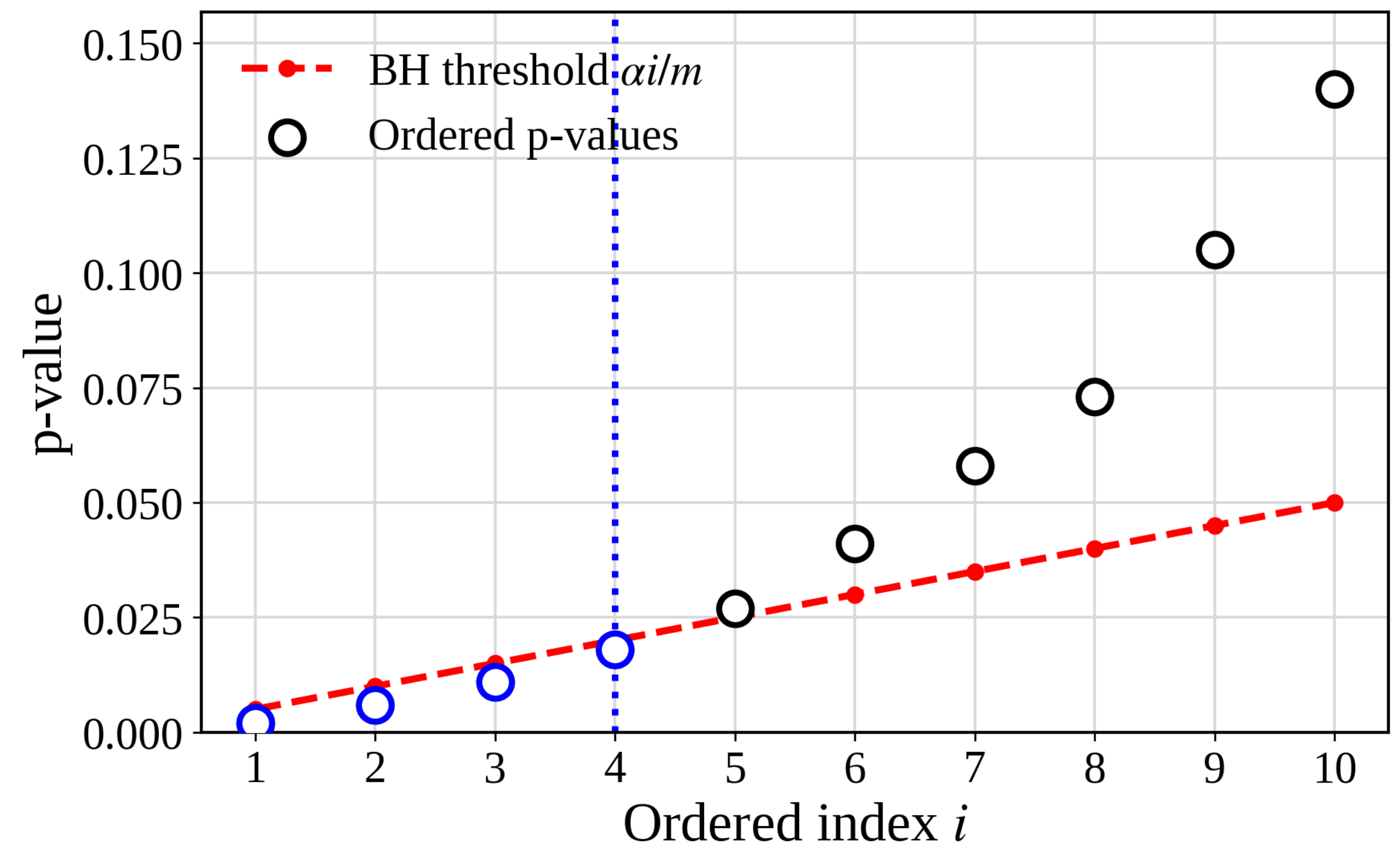


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- ▶ Order the p-values $p_{(1)} \leq p_{(2)} \leq \dots p_{(m)}$
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- ▶ Reject $H_{(j)}$ for all $j \leq j_0$



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Intuition of the BH procedure

$$\begin{aligned} j_0 &= \max \left\{ j : p_{(j)} \leq \frac{j}{m} \alpha \right\} \\ &= \max \left\{ j : p_{(j)} \leq \frac{\sum_i 1\{p_i \leq p_{(j)}\}}{m} \alpha \right\} \\ &= \max \left\{ j : \frac{mp_{(j)}}{\sum_i 1\{p_i \leq p_{(j)}\}} \leq \alpha \right\} \end{aligned}$$

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Interpretation: “finding the largest threshold for which the estimated FDP $\leq \alpha$ ”

FDR control guarantees of the BH procedure

Benjamini & Hochberg '95

If the p-values are mutually independent, then the BH procedure at α guarantees $\text{FDR} \leq \frac{m_0}{m} \alpha$

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If the p-values are mutually independent, then the BH procedure at α guarantees $\text{FDR} \leq \frac{m_0}{m} \alpha$

- ▶ The independence assumption can be relaxed to a type of positive dependence
 - *Positive Regression Dependency on a Subset (PRDS)*⁴
- ▶ Conservative if $m_0 \ll m$

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Improving upon the null proportion

- ▶ If we knew m_0 , would apply the BH at level $\alpha \times m/m_0$...
- ▶ In practice, no access to $m_0 \leadsto$ estimate?

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$$\Rightarrow \frac{m_0}{m} \approx \frac{2 \times \#\{j : p_j > 1/2\}}{m}$$

Storey-BH⁵

5. Storey, John D. "A direct approach to false discovery rates." Journal of the Royal Statistical Society Series B: Statistical Methodology (2002).

Storey-BH⁵

$$\widehat{\text{FDP}}^{\text{storey}}_{(t)} = \frac{1 + \#\{j : p_j > 1/2\}}{m/2} \cdot \frac{mt}{\#\{j : p_j \leq t\}}$$

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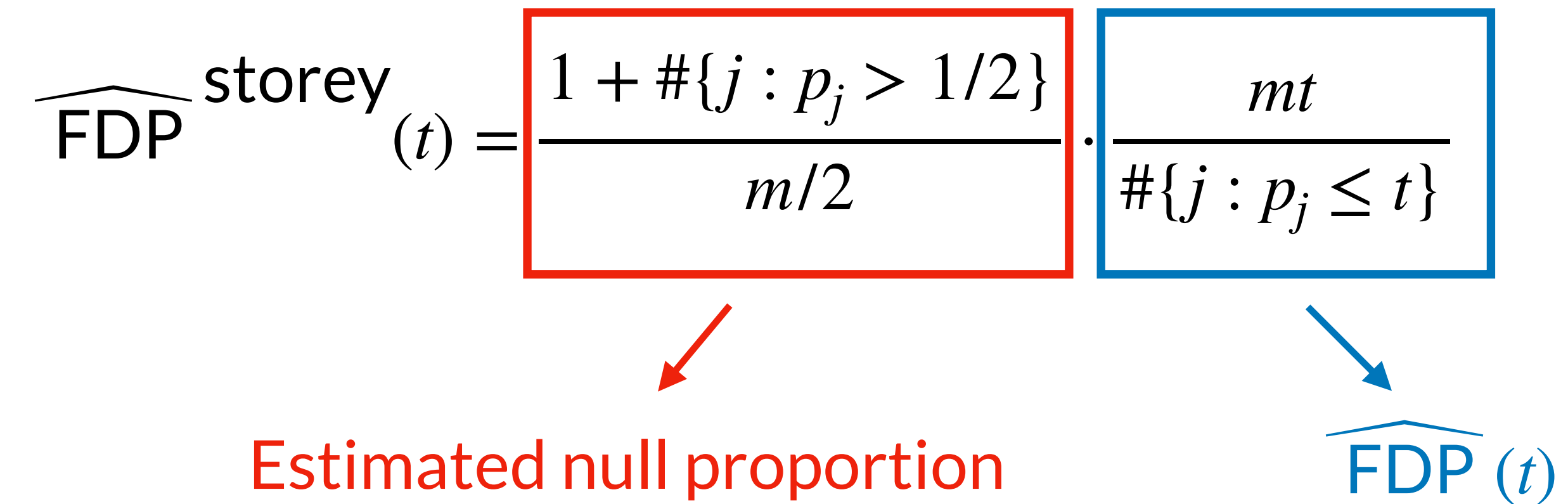
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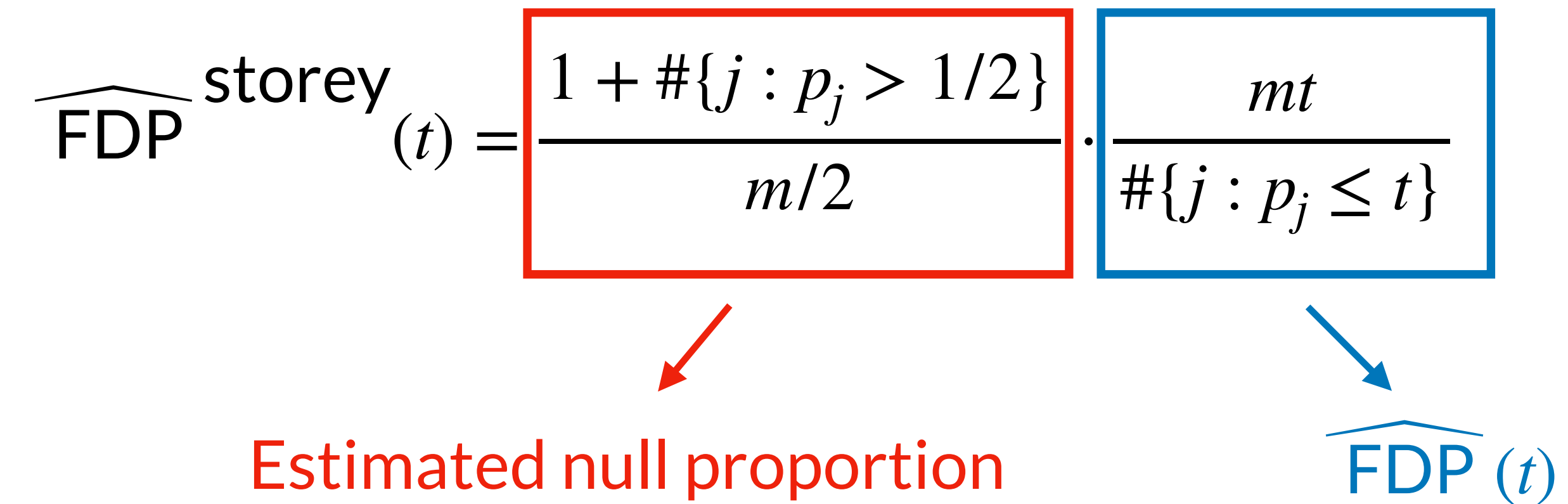
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Estimated null proportion $\widehat{\text{FDP}}(t)$

Let $\tau = \max\{t \leq 1/2 : \widehat{\text{FDP}}(t) \leq \alpha\}$ and $\mathcal{R} = \{j : p_j \leq \tau\}$

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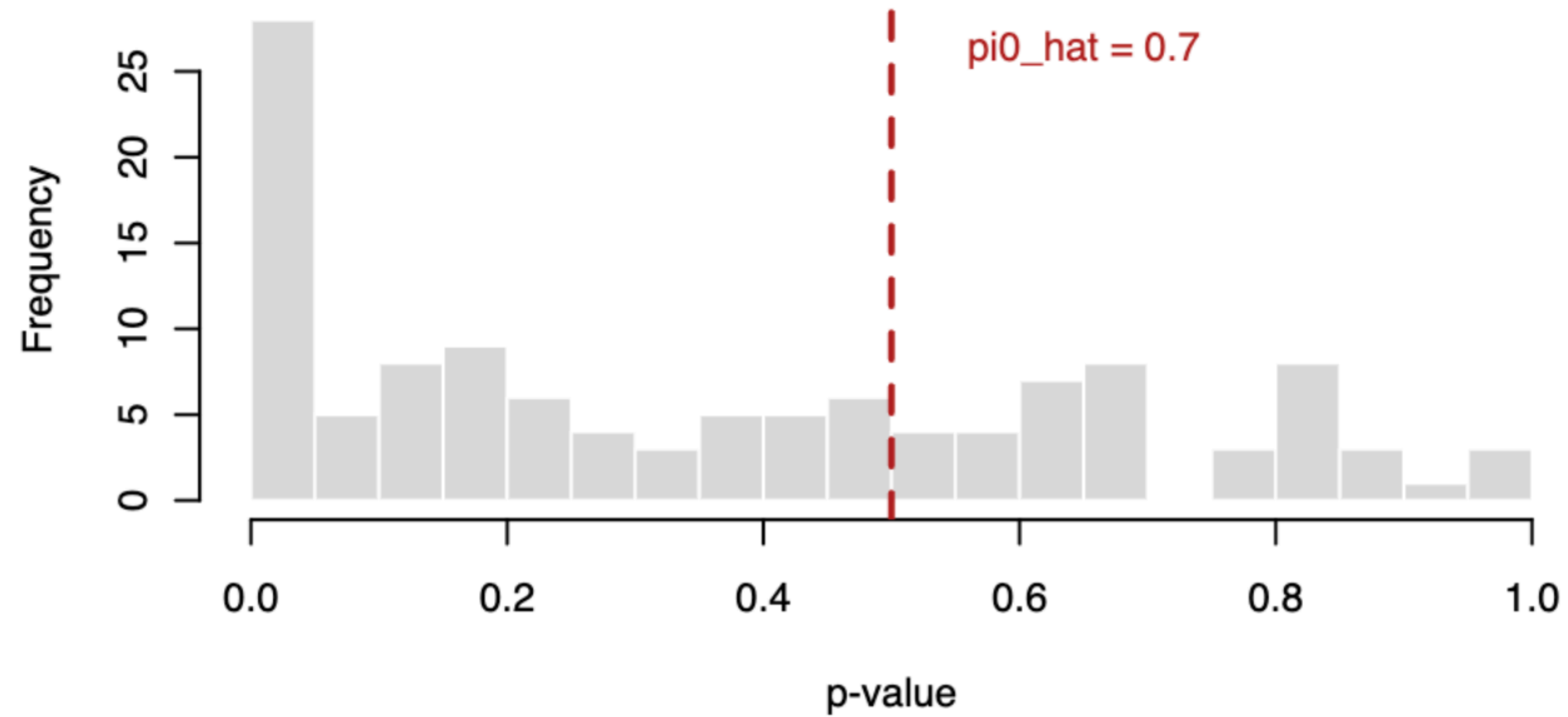
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What if the p-values are not independent?

A negative result

There are joint distributions of p-values for which FDR of the BH procedure at level α is at least $\min(\alpha \cdot S(m), 1)$, where

$$S(m) = 1 + \frac{1}{2} + \dots + \frac{1}{m}$$

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Grows at the logarithmic rate!

FDR control guarantees of the BH procedure

What if the p-values are not independent?

A positive result

Under dependence, the BH procedure at level α satisfies $\text{FDR} \leq \alpha S(m) \frac{m_0}{m}$

FDR control guarantees of the BH procedure: a summary

	BH	storey-BH
Indep/PRDS	$\frac{m_0}{m}\alpha$	α
Arbitrary	$\frac{m_0}{m}\alpha S(m)$	

FDR control guarantees of the BH procedure: a summary

	BH	storey-BH
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- ▶ Empirically, BH is robust to dependence in terms of **controlling FDR**
- ▶ But **Var(FDP)** can blow up under dependence

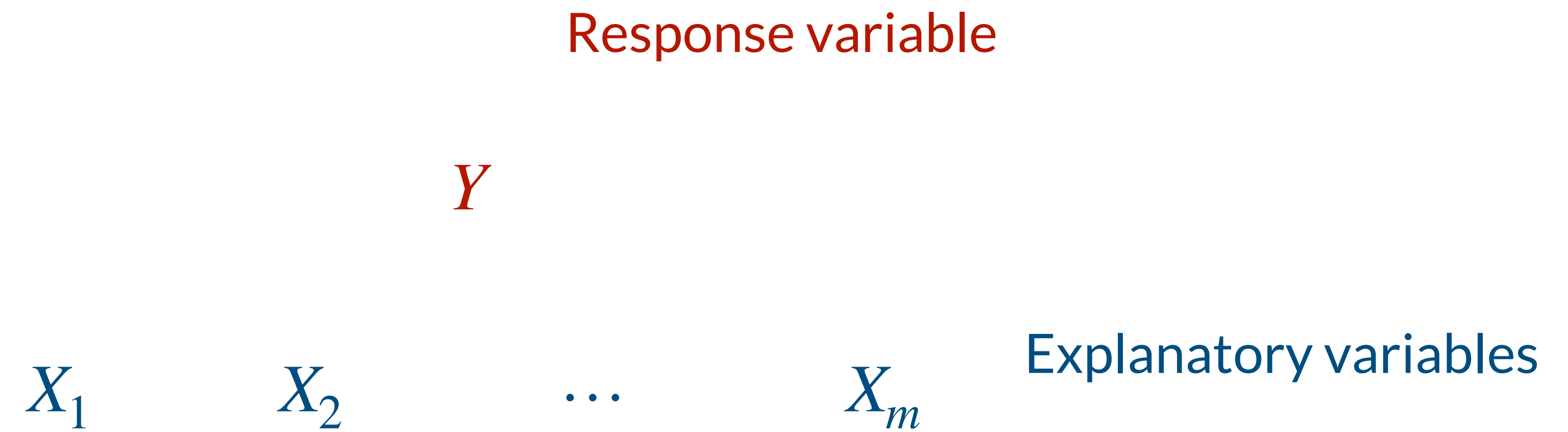
The feature selection problem

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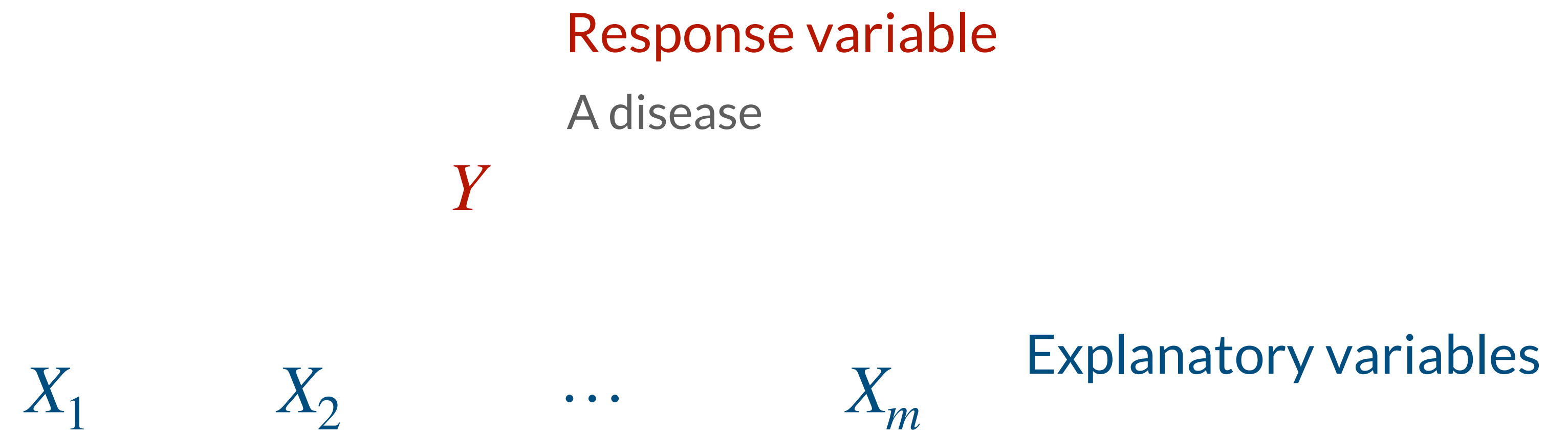
Response variable

Y

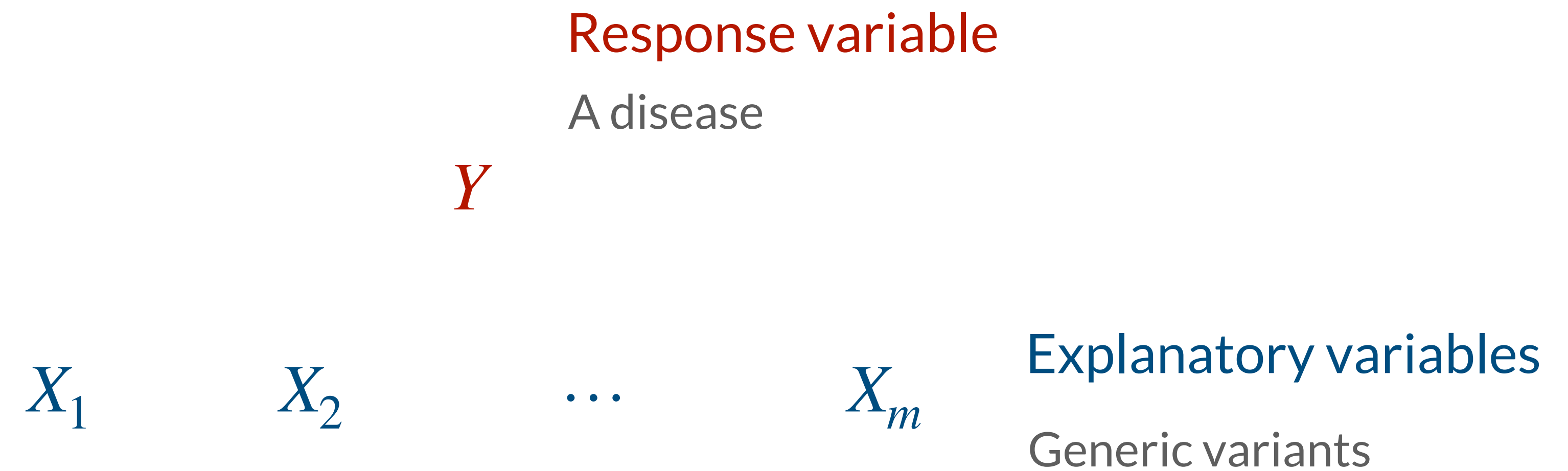
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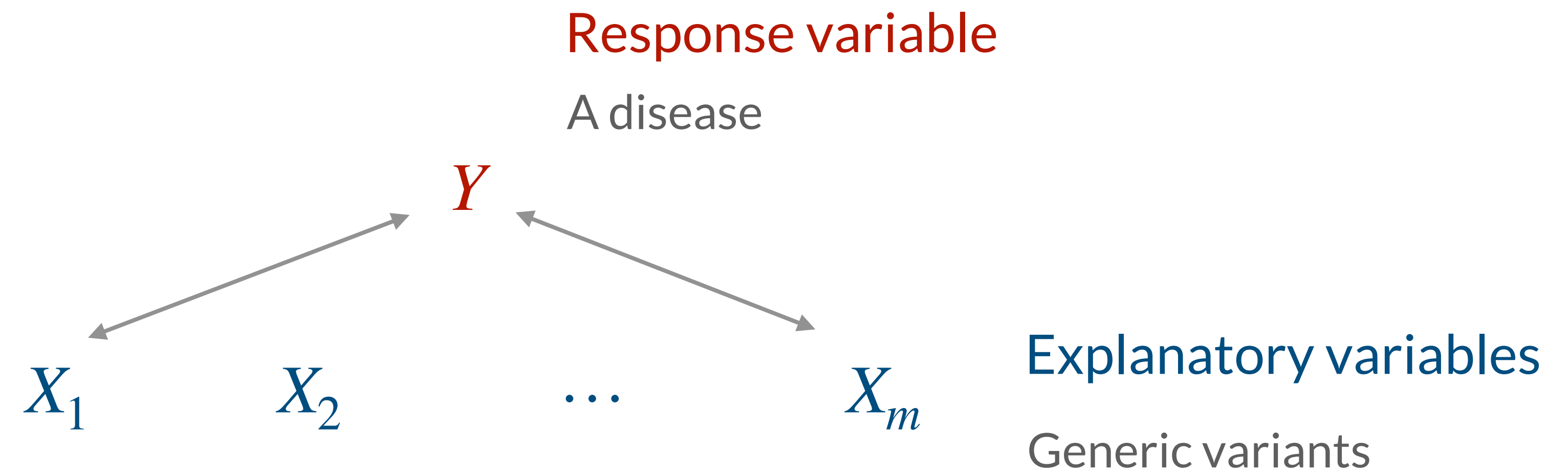
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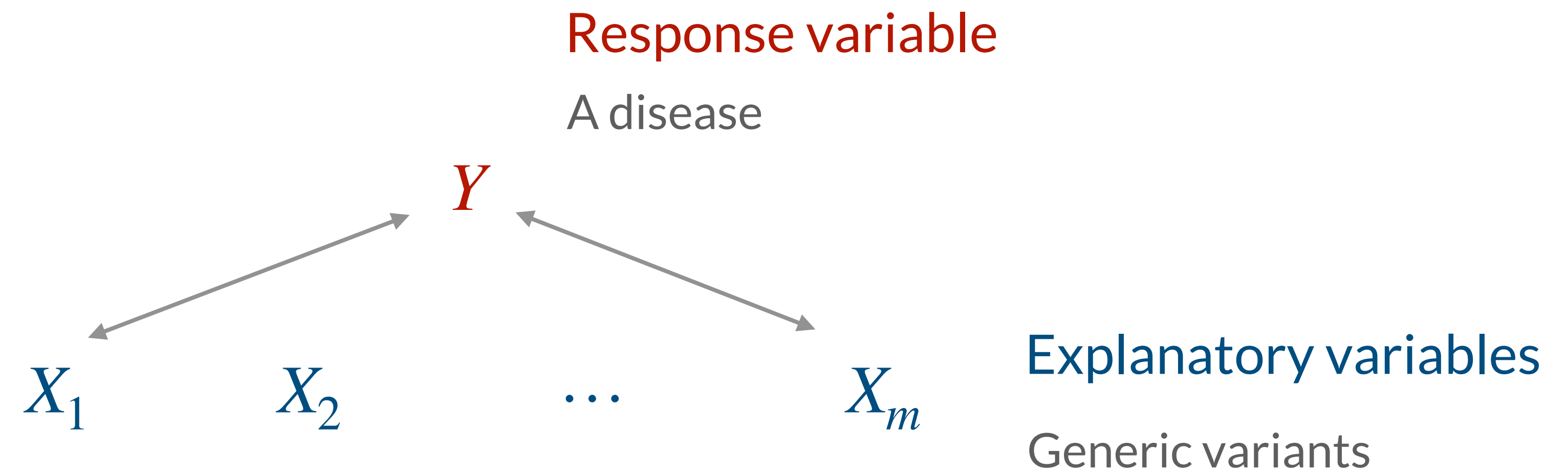
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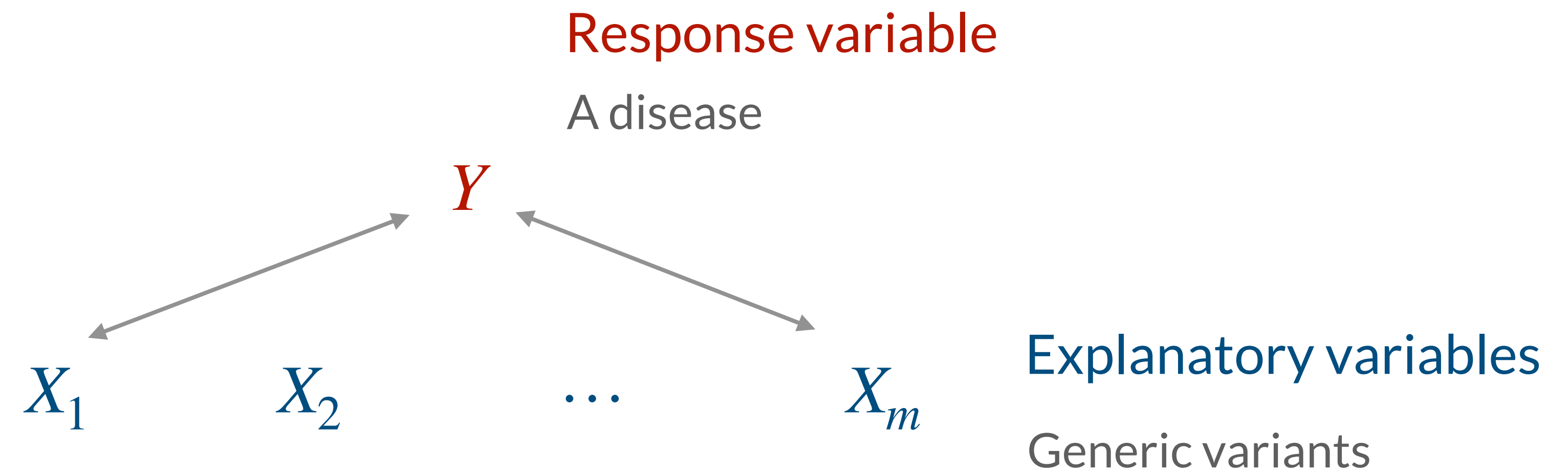


The feature selection problem



$$Y = \sum_{j=1}^m \beta_j X_j + \epsilon, \quad \epsilon \sim N(0, \sigma^2) \quad H_j : \beta_j = 0$$

The feature selection problem



$$Y = \sum_{j=1}^m \beta_j X_j + \epsilon, \quad \epsilon \sim N(0, \sigma^2)$$

$$H_j : \beta_j = 0$$

MT problem; aim at FDR control

Controlled feature selection as a MT problem

$$Y = \sum_{j=1}^m \beta_j X_j + \epsilon, \quad \epsilon \sim N(0, \sigma^2) \quad H_j : \beta_j = 0$$

Option:

- ▶ Run linear regression
- ▶ Obtain a two-sided p-value p_j from the t test
- ▶ Apply BH to $\{p_j\}$ at the target FDR level

Controlled feature selection as a MT problem

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Lasso, random forest, neural networks ...
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Option:

- ▶ Run linear regression  May want to use more complicated regression methods
Lasso, random forest, neural networks ...
- ▶ Obtain a two-sided p-value p_j from the t test  P-values may be dependent
- ▶ Apply BH to $\{p_j\}$ at the target FDR level

The knockoffs method⁶

6. Barber, Rina Foygel, and Emmanuel J. Candès. "Controlling the false discovery rate via knockoffs." *Annals of Statistics* (2015).

7. Candès, Emmanuel, et al. "Panning for gold: 'model-X' knockoffs for high dimensional controlled variable selection." *Journal of the Royal Statistical Society Series B: Statistical Methodology* (2018).

The knockoffs method⁶

- ▶ This course: **fixed-X** setting

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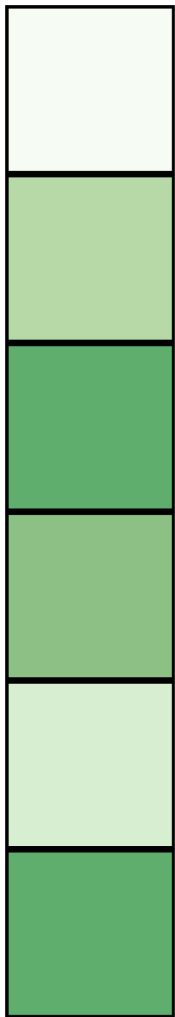
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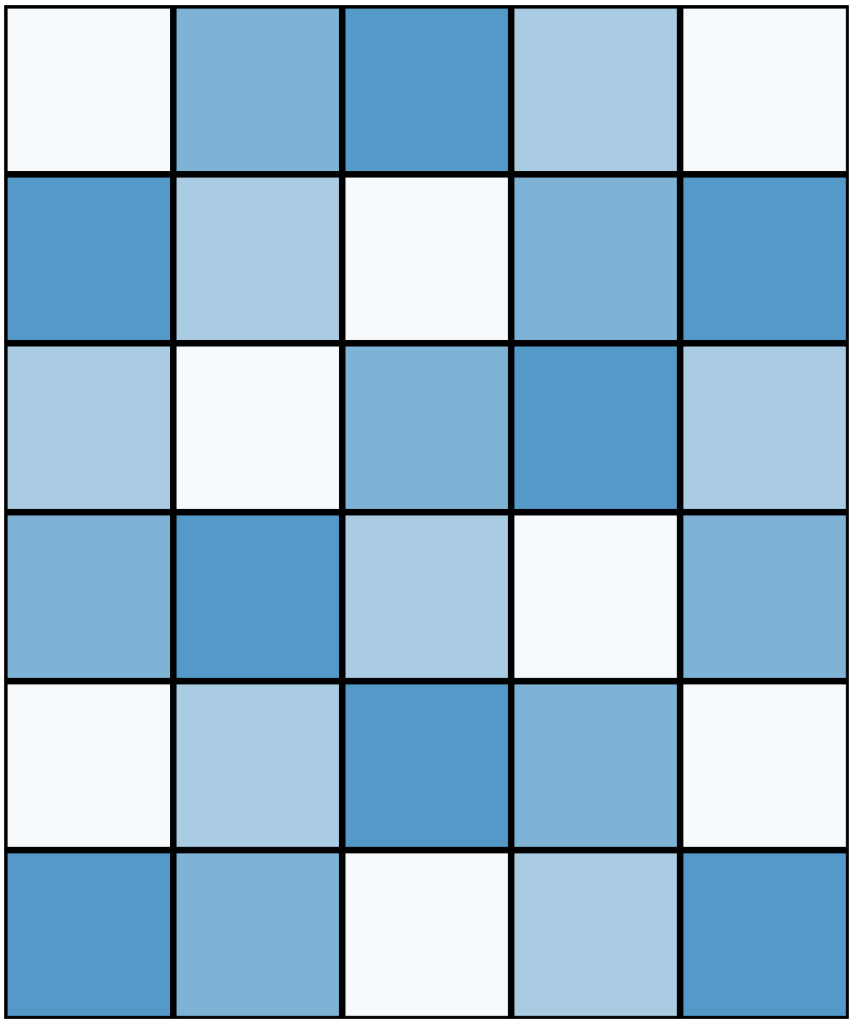
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The knockoffs method

Response Y

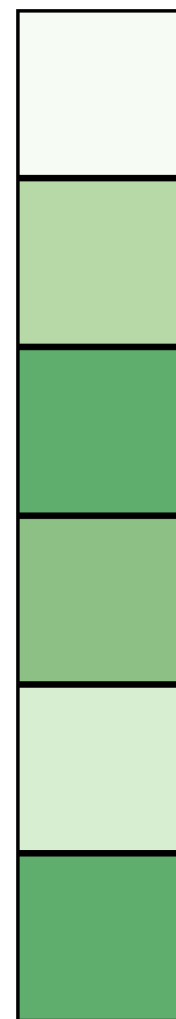


Design matrix X

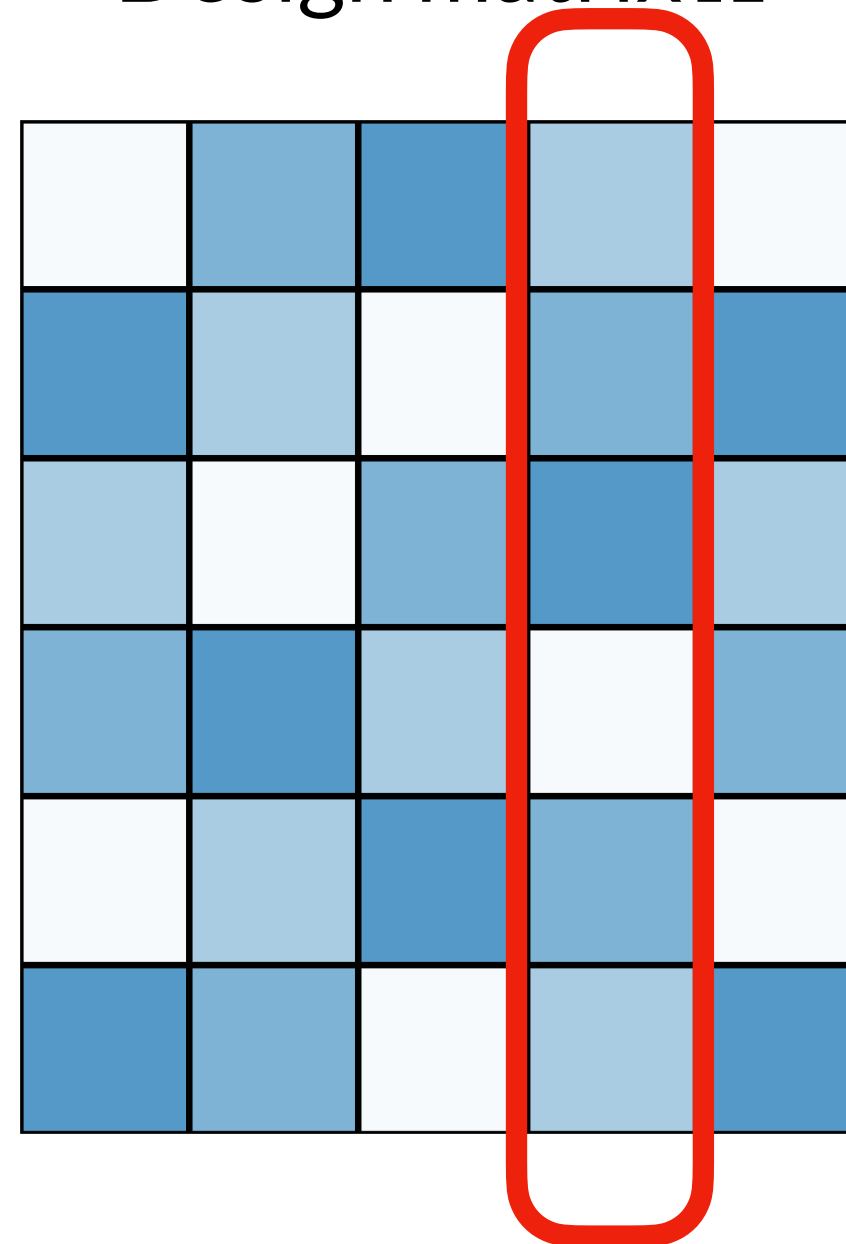


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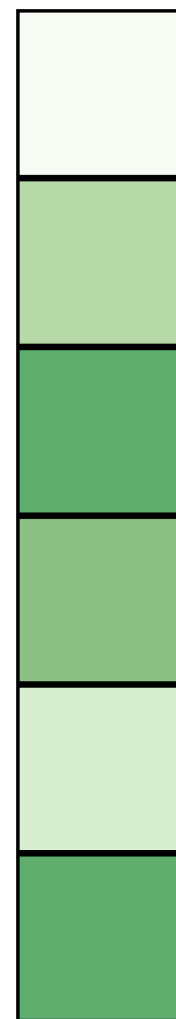


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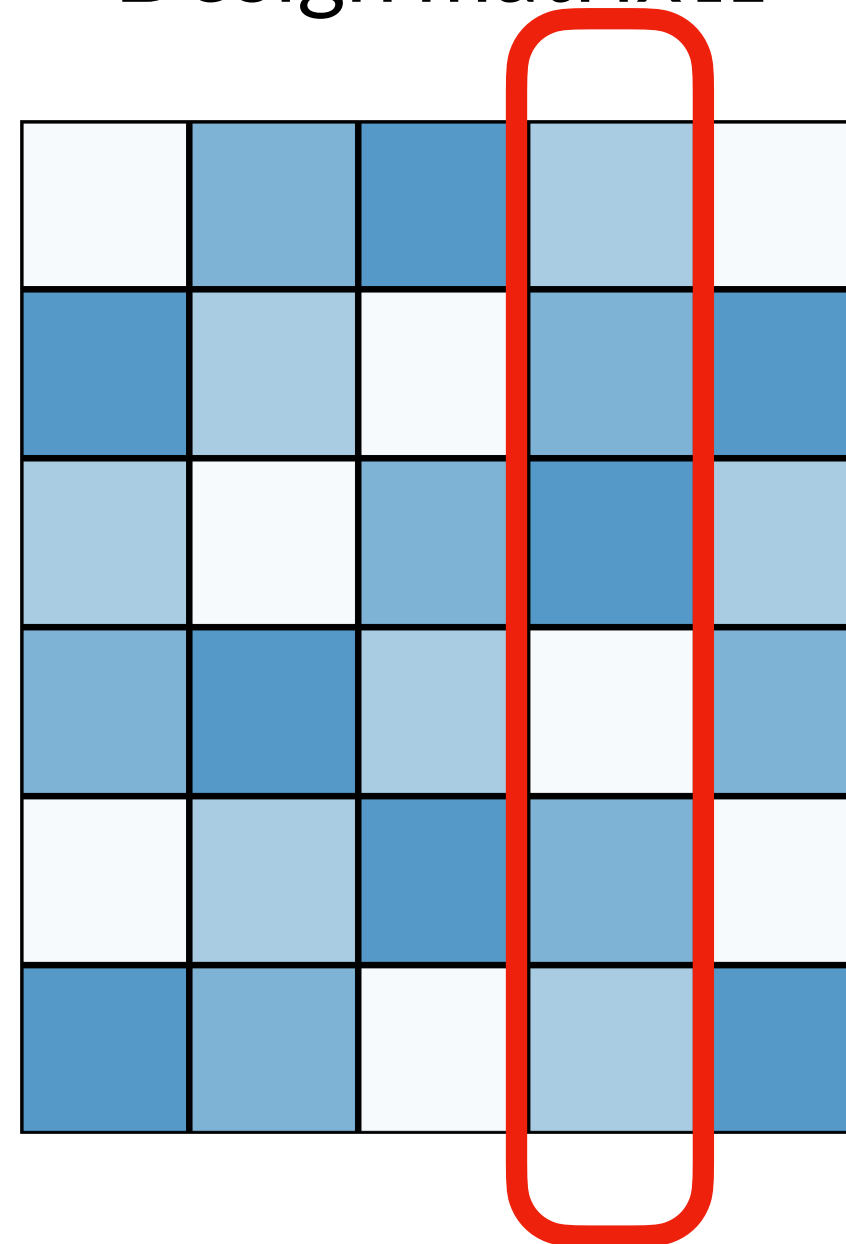


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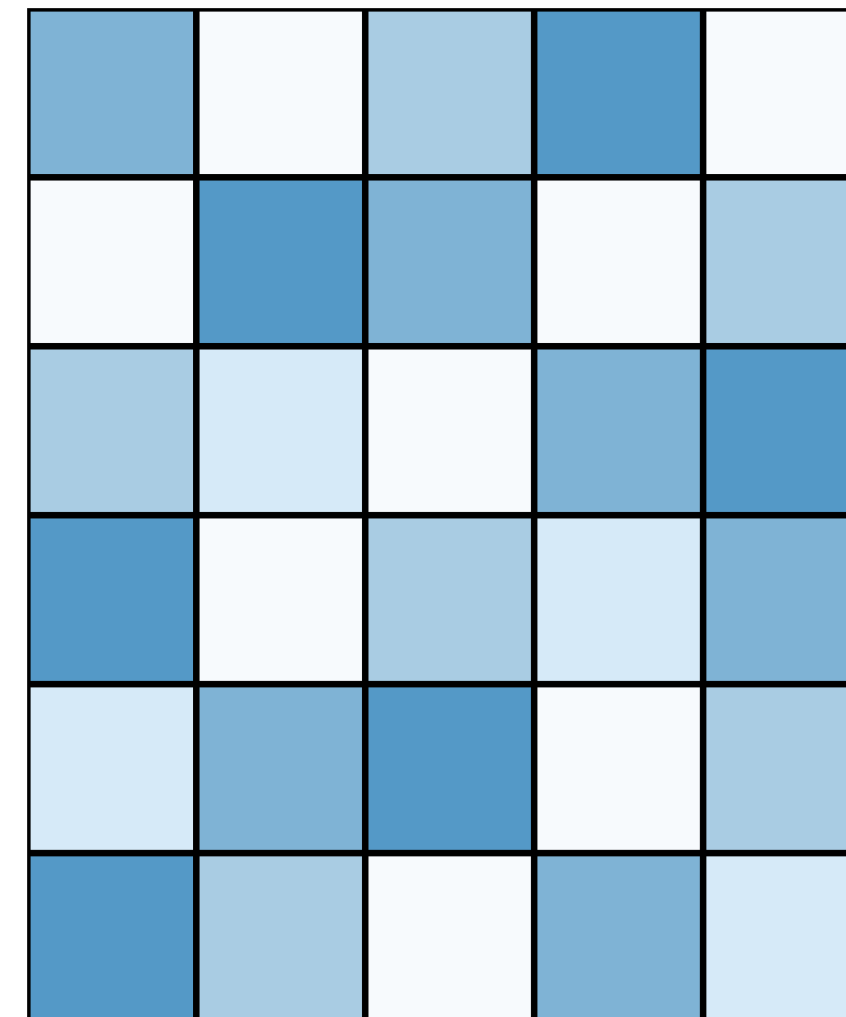
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Design matrix X



Knockoff matrix $\tilde{X}$

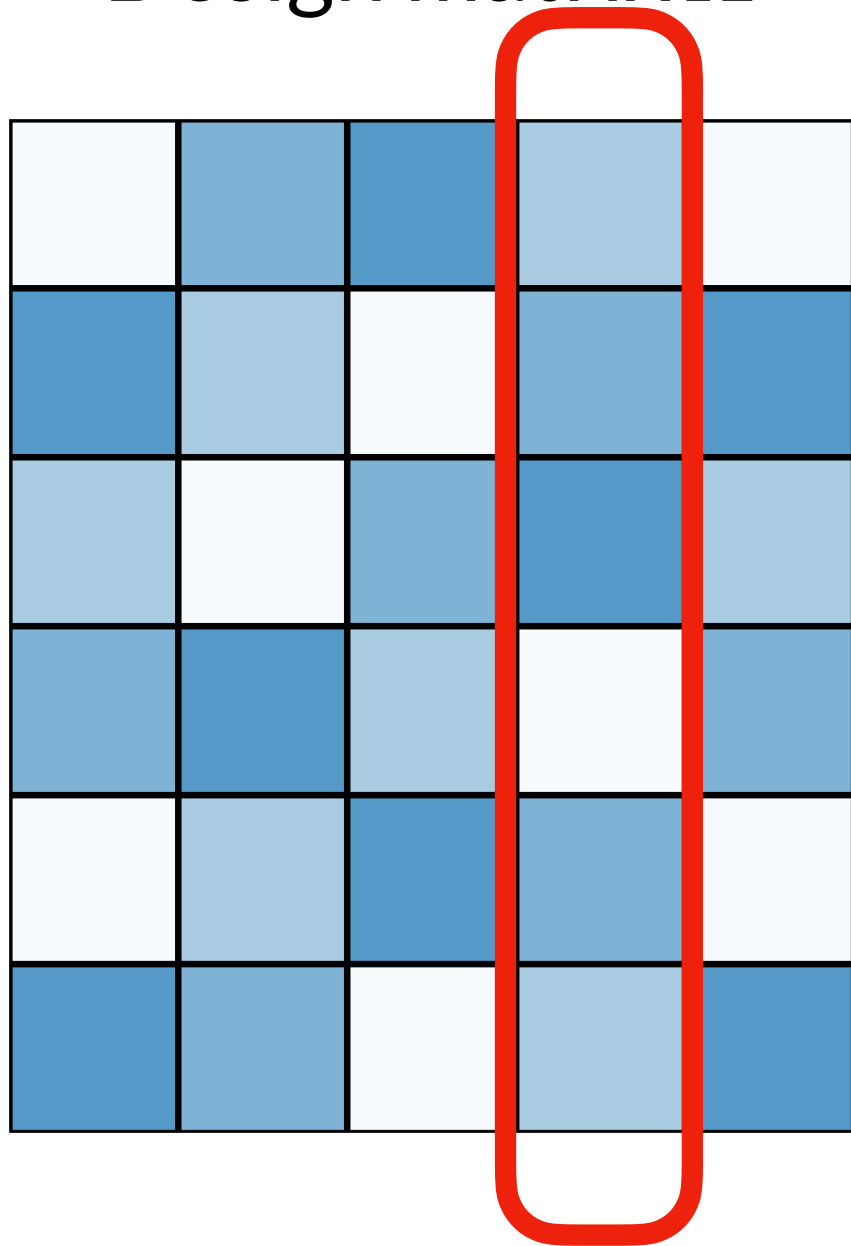


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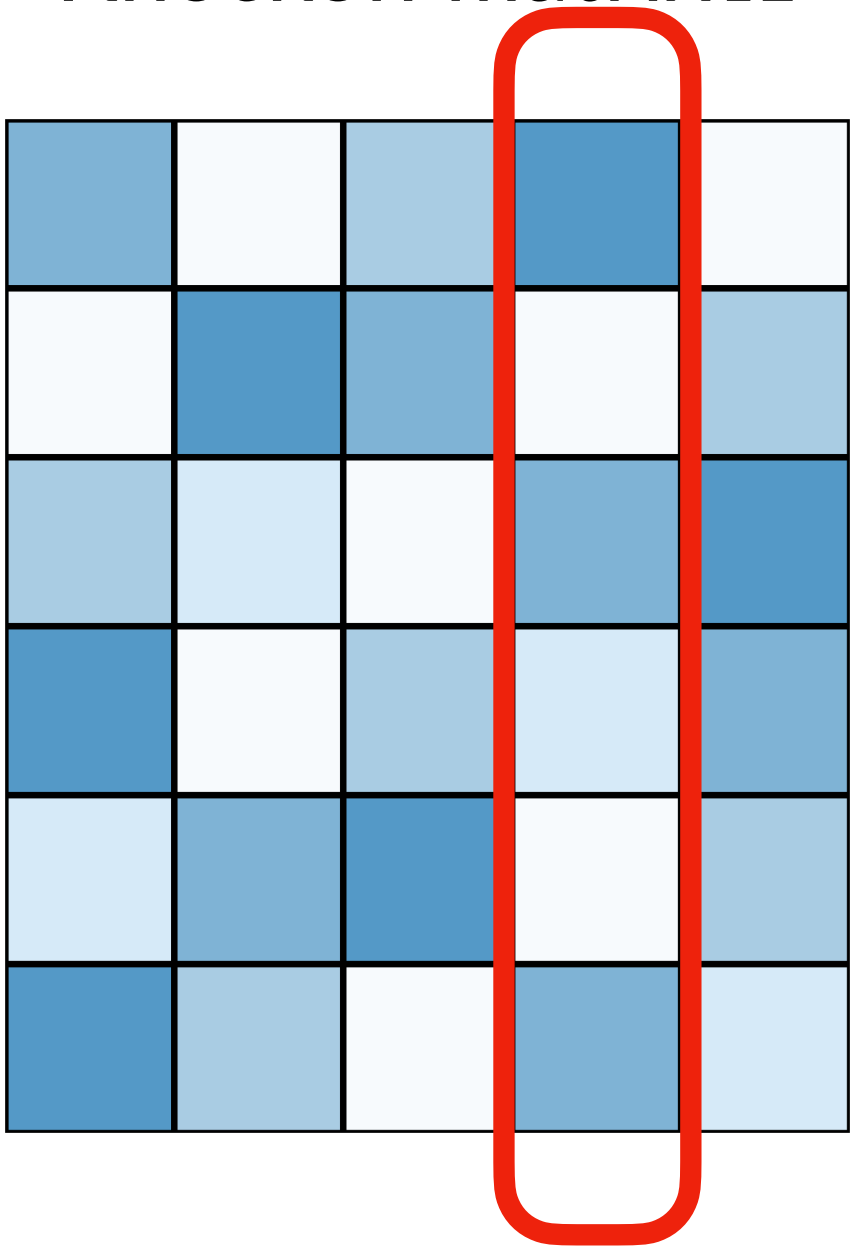
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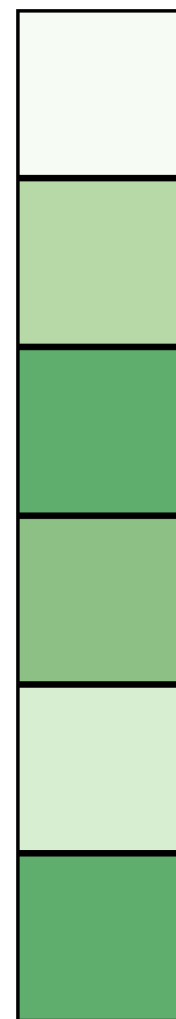


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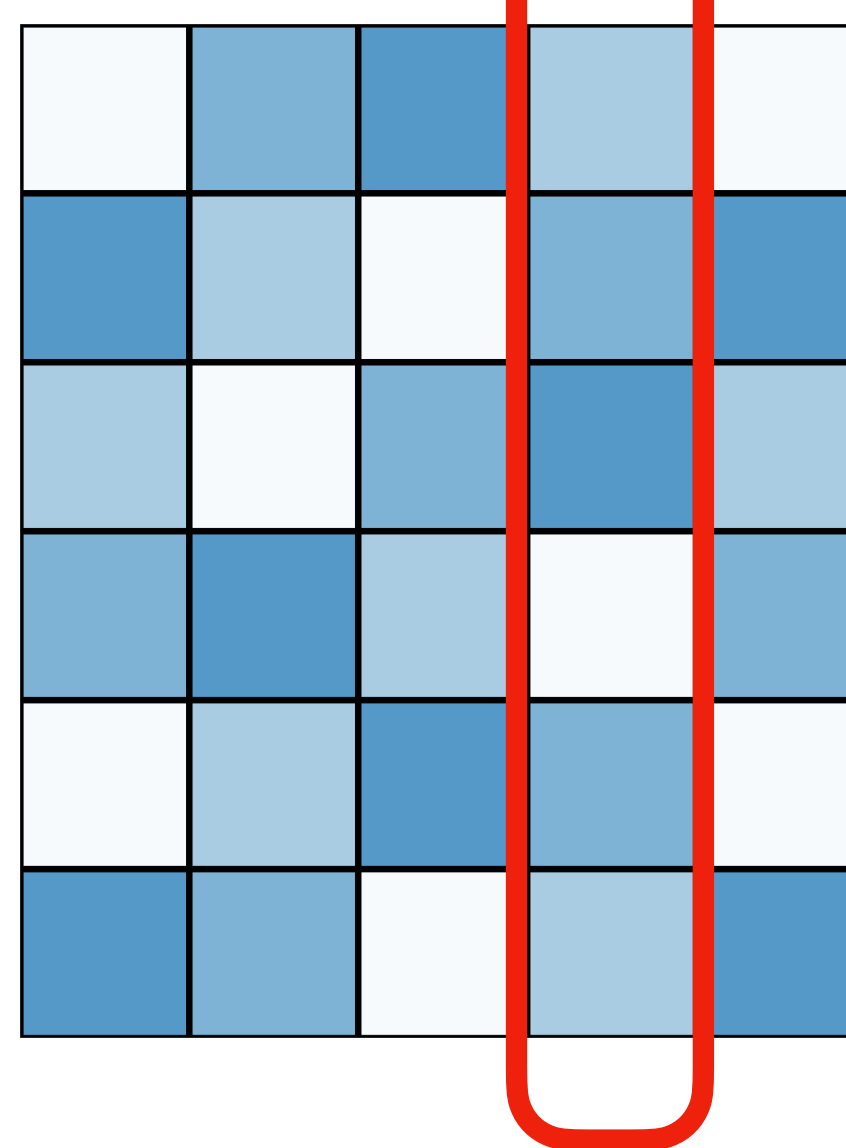


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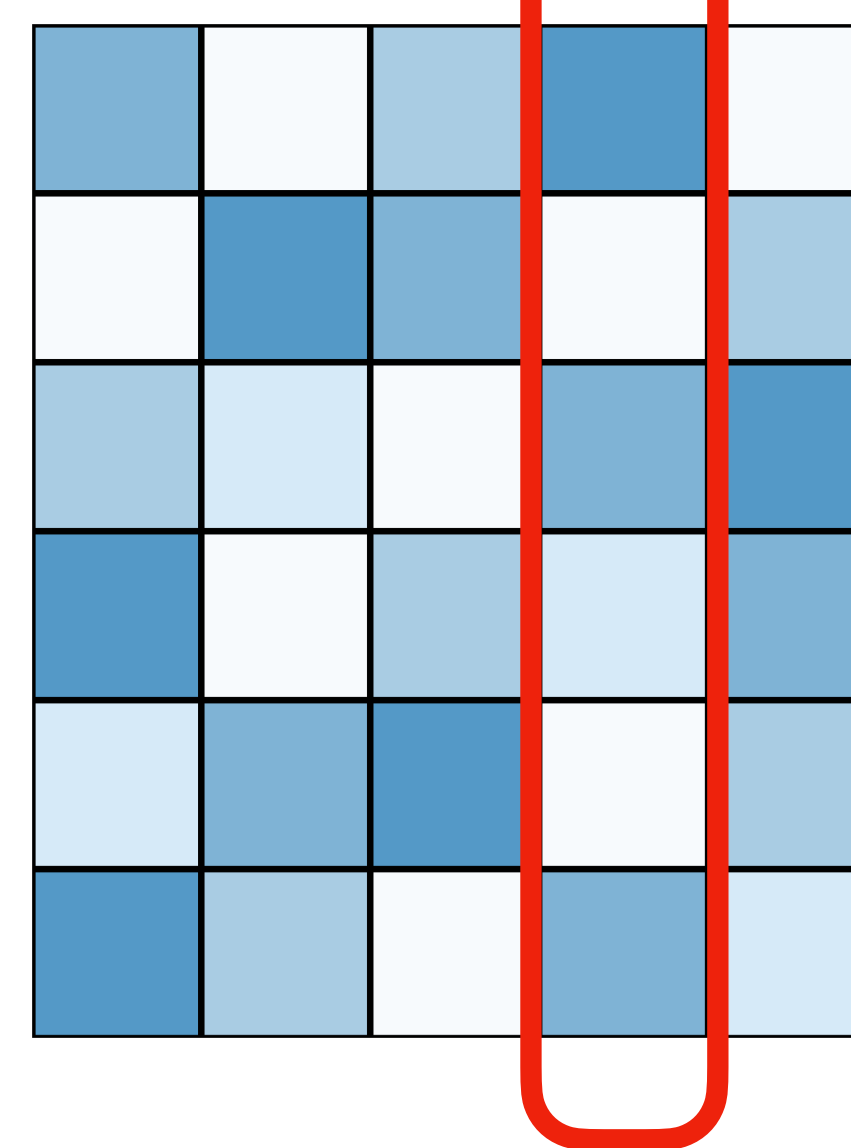
Response Y



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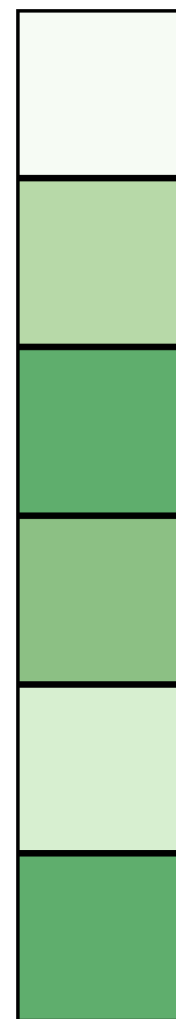


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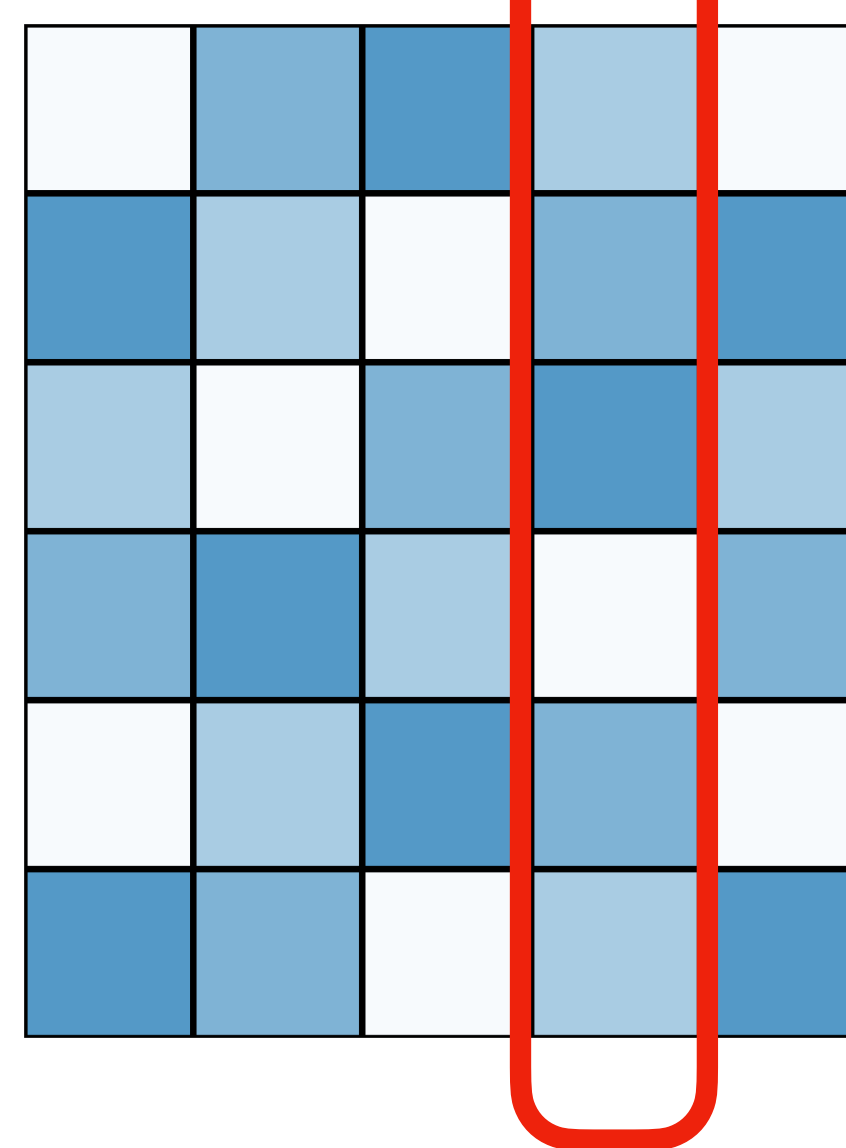


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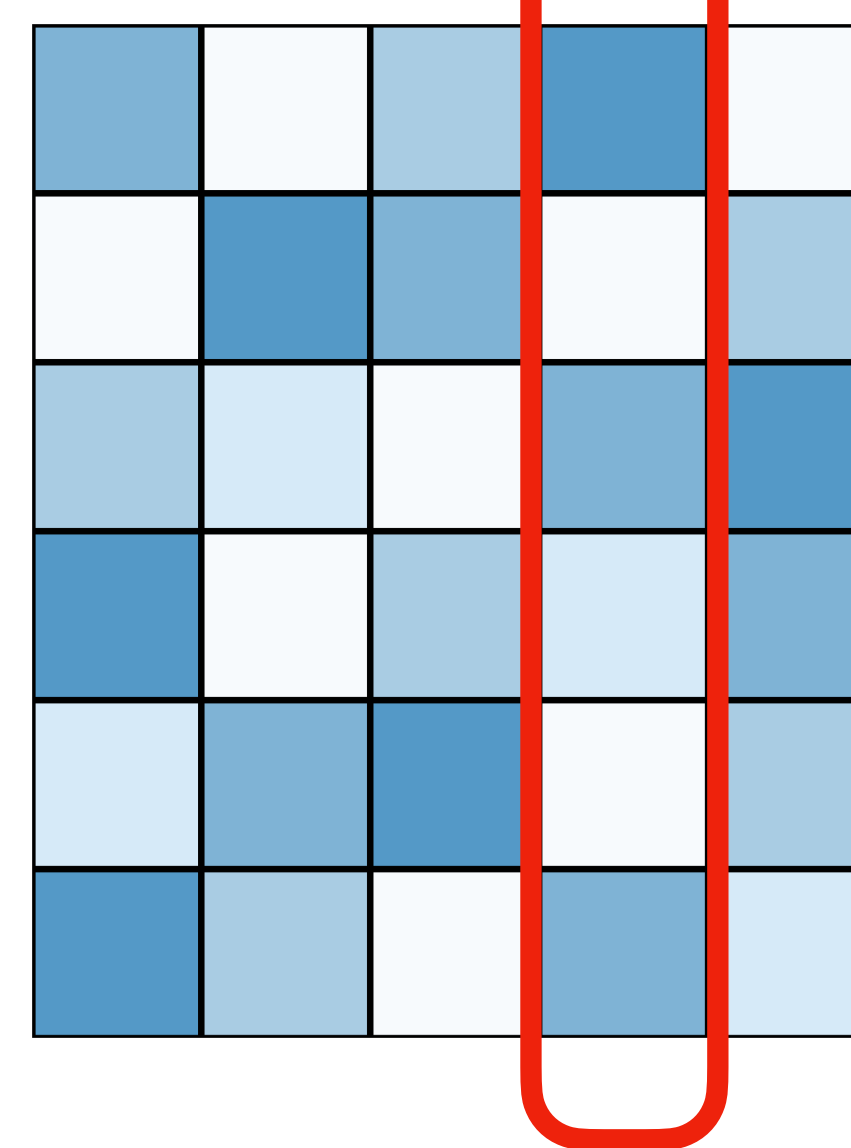
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Design matrix X



Knockoff matrix $\tilde{X}$



“Select j if X_j more significant than $\tilde{X}_j$.”

Knockoff construction

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- ▶ Given the design matrix $X \in \mathbb{R}^{n \times m}$

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- ▶ Choose $U \in \mathbb{R}^{n \times m}$ with orthonormal columns such that $U \perp X$
- ▶ Let $\tilde{X} = X(I - (X^\top X)^{-1}D) + U \cdot (2D - D(X^\top X)^{-1}D)^{1/2}$

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Want D_j to be large to contrast X_j and $\tilde{X}_j$
e.g., $\max \sum_i D_{ii}$ subject to $D \preceq 2(X^\top X)$

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Compute feature importance statistics

Feature importance statistics $W = \mathcal{W}(X, \tilde{X}, Y) \in \mathbb{R}^m$

Requirement

- ▶ **Antisymmetry:** if we swap X_j with $\tilde{X}_j$, W_j becomes $-W_j$, while all other W_k remains unchanged
- ▶ **Sufficiency:** $\mathcal{W}(\cdot)$ can be written as a function of $[X, \tilde{X}]^\top [X, \tilde{X}]$ and $[X, \tilde{X}]^\top Y$

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Examples

- ▶ $W_j = |X_j^\top Y| - |\tilde{X}_j^\top Y|$

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- ▶ $W_j = \lambda_j - \lambda_{j+d}$, where $\lambda_k = \sup\{\lambda : \beta_k \neq 0\}$

Rejection set

Form of rejection set: $\{j : W_j \geq t\}$

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Need to pick t

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- For a candidate t , construct FDP estimate

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$$\widehat{\text{FDP}}(t) = \frac{1 + \boxed{\sum_j 1\{W_j \leq -t\}}}{\sum_j 1\{W_j \geq t\}} = \sum_{j \text{ nonnull}} 1\{W_j \leq -t\} + \sum_{j \text{ null}} 1\{W_j \leq -t\} \\ \approx 0 + \sum_{j \text{ null}} 1\{W_j \geq t\}$$

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- Reject $\mathcal{R}(T) = \{j : W_j \geq T\}$

FDR control guarantees of the knockoffs procedure

Barber and Candès '15

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FDR control guarantees of the knockoffs procedure

Barber and Candès '15

The knockoffs procedure applied at level α guarantees $\text{FDR} \leq \alpha$

- ▶ Accommodates correlation among features
- ▶ Allows for using complex models
- ▶ No m_0/m factor
- ▶ No use of p-values
- ▶ Can be conservative when the number of signals is low

Lecture 4: Adaptive/interactive inference

Structure in hypothesis testing

Hypotheses: $H_1, H_2, \dots, H_m$

Structure in hypothesis testing

Hypotheses: $H_1, H_2, \dots, H_m$

Side information: $x_1, x_2, \dots, x_m$

Structure in hypothesis testing

Hypotheses: $H_1, H_2, \dots, H_m$

Side information: $x_1, x_2, \dots, x_m$



Encode prior knowledge, location, correlation with others ...

Structure in hypothesis testing

GWAS

H_j : SNP _{j} is not associated w/ the phenotype

Structure in hypothesis testing

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- ▶ Association w/ other phenotypes
- ▶ Functional annotation
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fMRI: testing for activation at a voxel

H_j : voxel_j is not activated

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- ▶ Similarity to other compounds
- ▶ ...

fMRI: testing for activation at a voxel

H_j : voxel_{*j*} is not activated

- ▶ Anatomical region
- ▶ Neighboring voxels
- ▶ Tissue type
- ▶ ...

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e.g., a Bayesian may impose a model and learn $\mathbb{P}(H_j = 1 \mid p_j = p, X_j = x)$

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e.g., a Bayesian may impose a model and learn $\mathbb{P}(H_j = 1 \mid p_j = p, X_j = x)$

But...

- ▶ May subject to model misspecification
- ▶ Use the same data for modeling and inference $\leadsto$ frequentist guarantee violation

Modeling the structure

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- ▶ Challenge: how to model the structure?

Option 2:

- ▶ use the a subset of the dataset to learn the structure
- ▶ conduct inference on the other half

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But...

- ▶ Need to determine the splitting ratio a priori
- ▶ Generally a loss of efficiency

Interactive data analysis with FDR control⁸

8. Lei, Lihua, and William Fithian. "AdaPT: an interactive procedure for multiple testing with side information." *Journal of the Royal Statistical Society Series B: Statistical Methodology* (2018).

Interactive data analysis with FDR control⁸

Idea

- gradually reveal information for modeling
- use the unrevealed information for inference

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Interactive data analysis with FDR control⁸

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- use the unrevealed information for inference

Setup

- $p_1, \dots, p_m \mid x_1, \dots, x_m$ are independent

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Interactive data analysis with FDR control⁸

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- gradually reveal information for modeling
- use the unrevealed information for inference

Setup

- $p_1, \dots, p_m \mid x_1, \dots, x_m$ are independent

AdaPT

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Use the threshold functions to encode the structure

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- ▶ Go down the list of thresholds until an FDP estimate $\leq \alpha$
- ▶ Use a **mirror process** to estimate the number of false discoveries

Interactive data analysis with FDR control

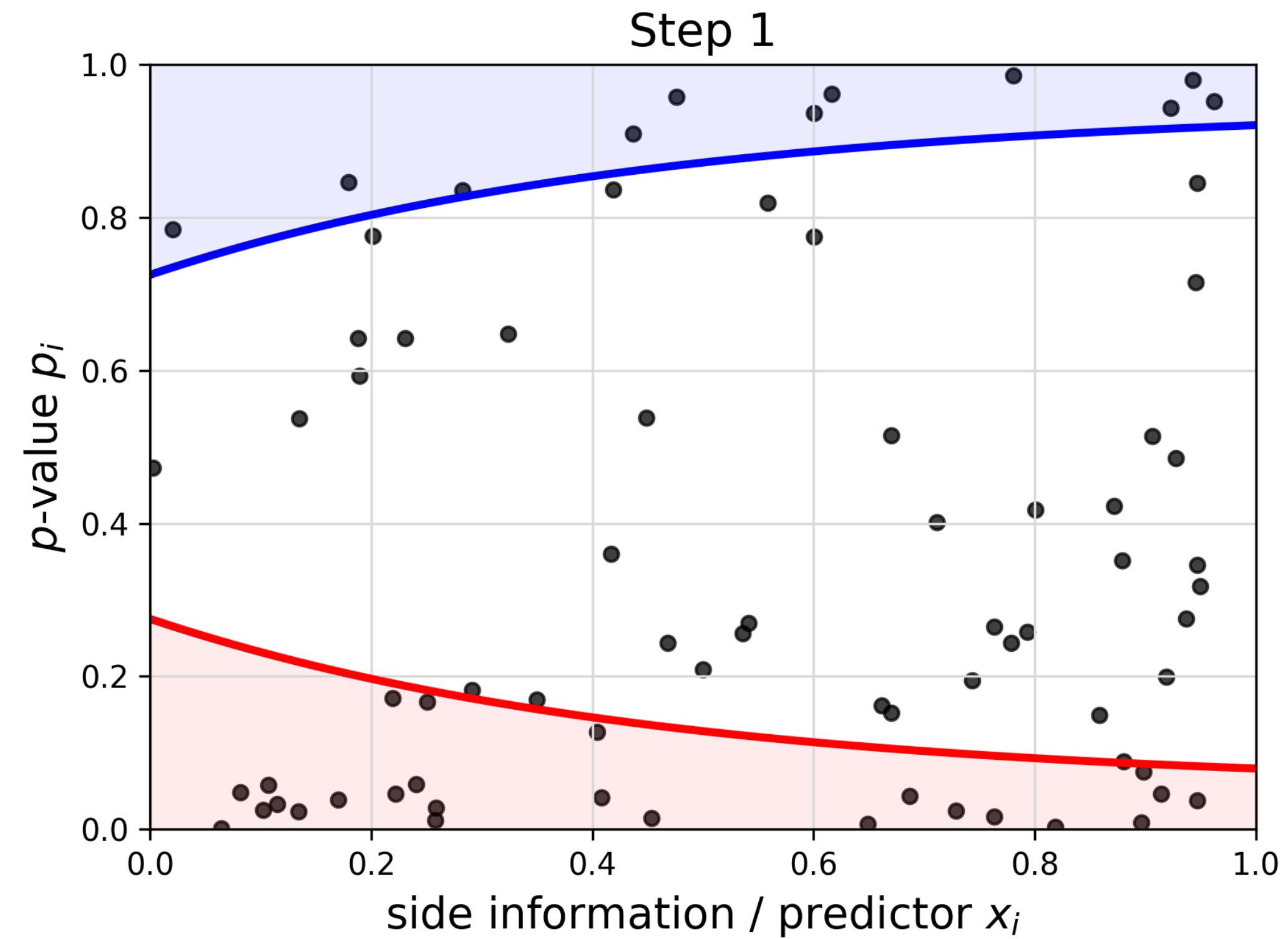
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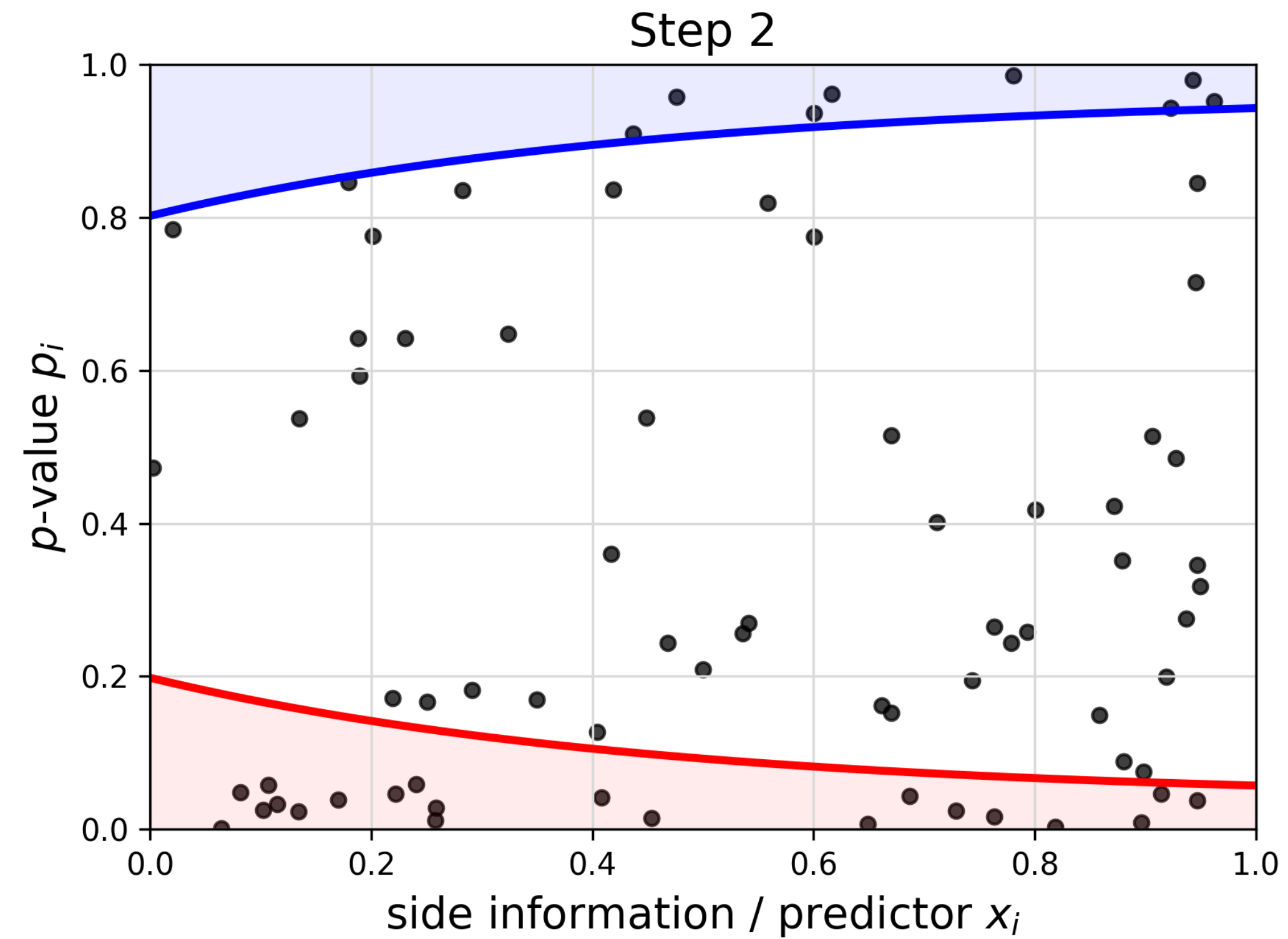
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$$\#\{j \text{ null} : p_j \leq s_t(x_j)\} \approx \#\{j : p_j \geq 1 - s_t(x_j)\}$$

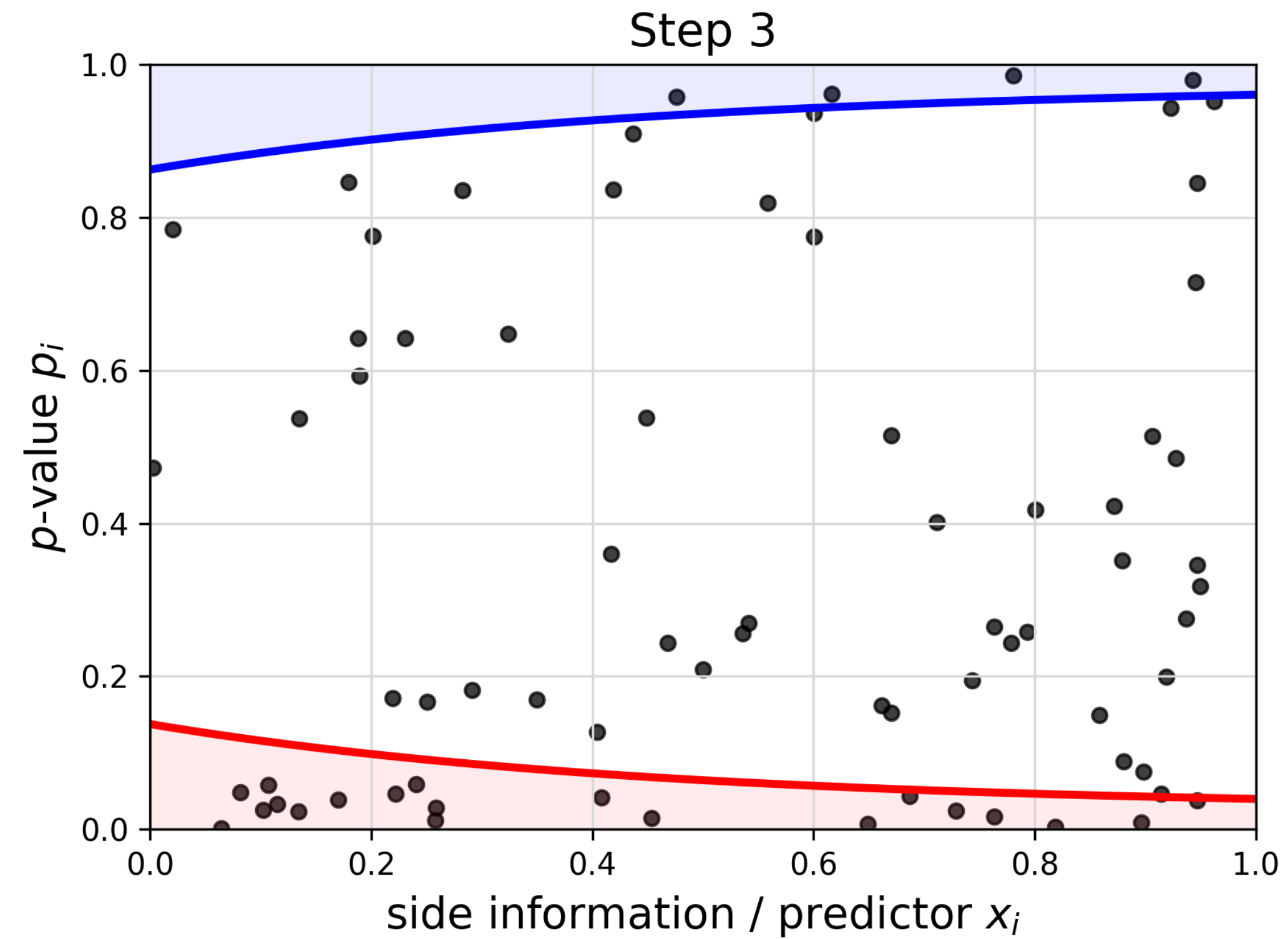
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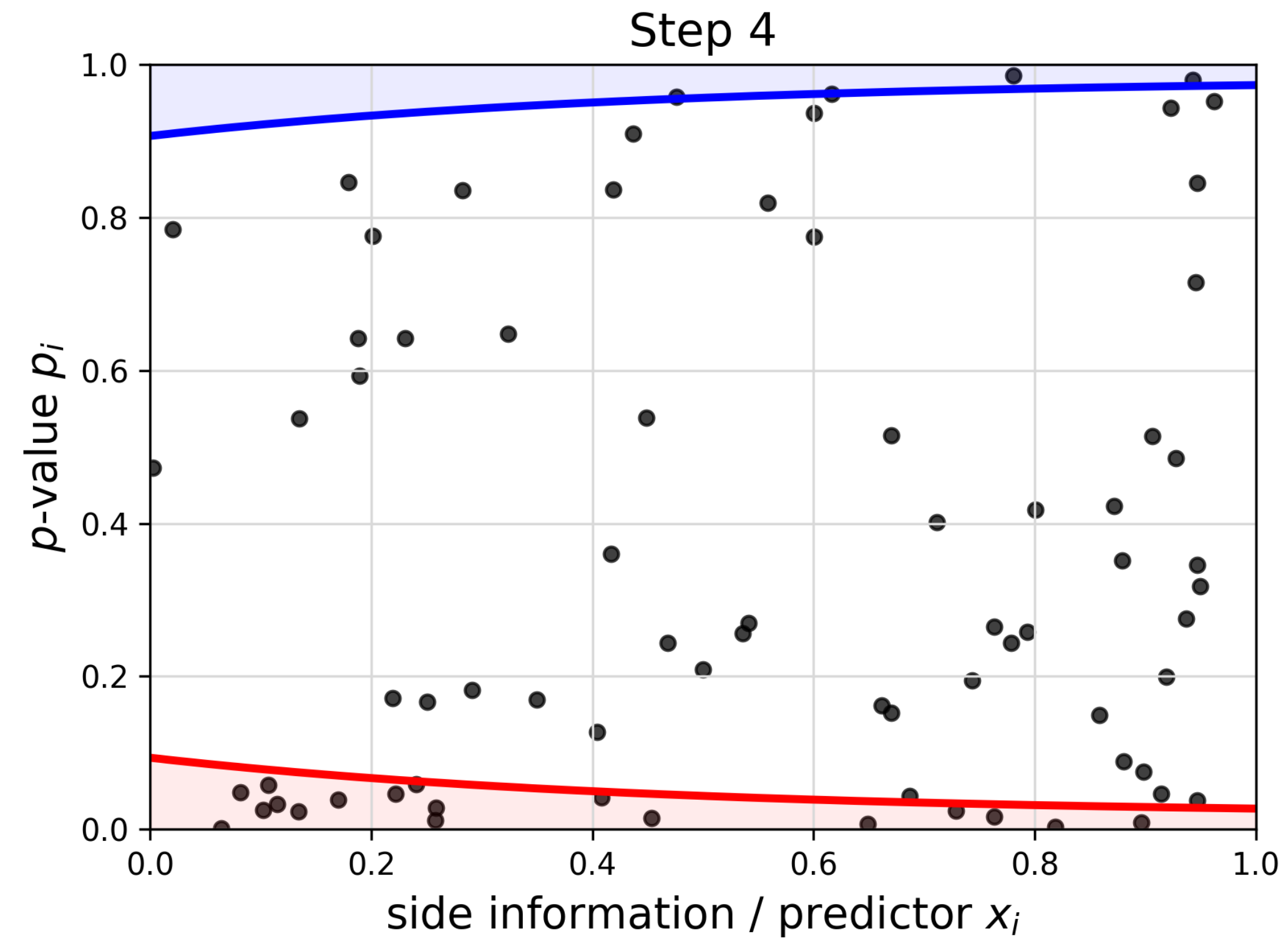
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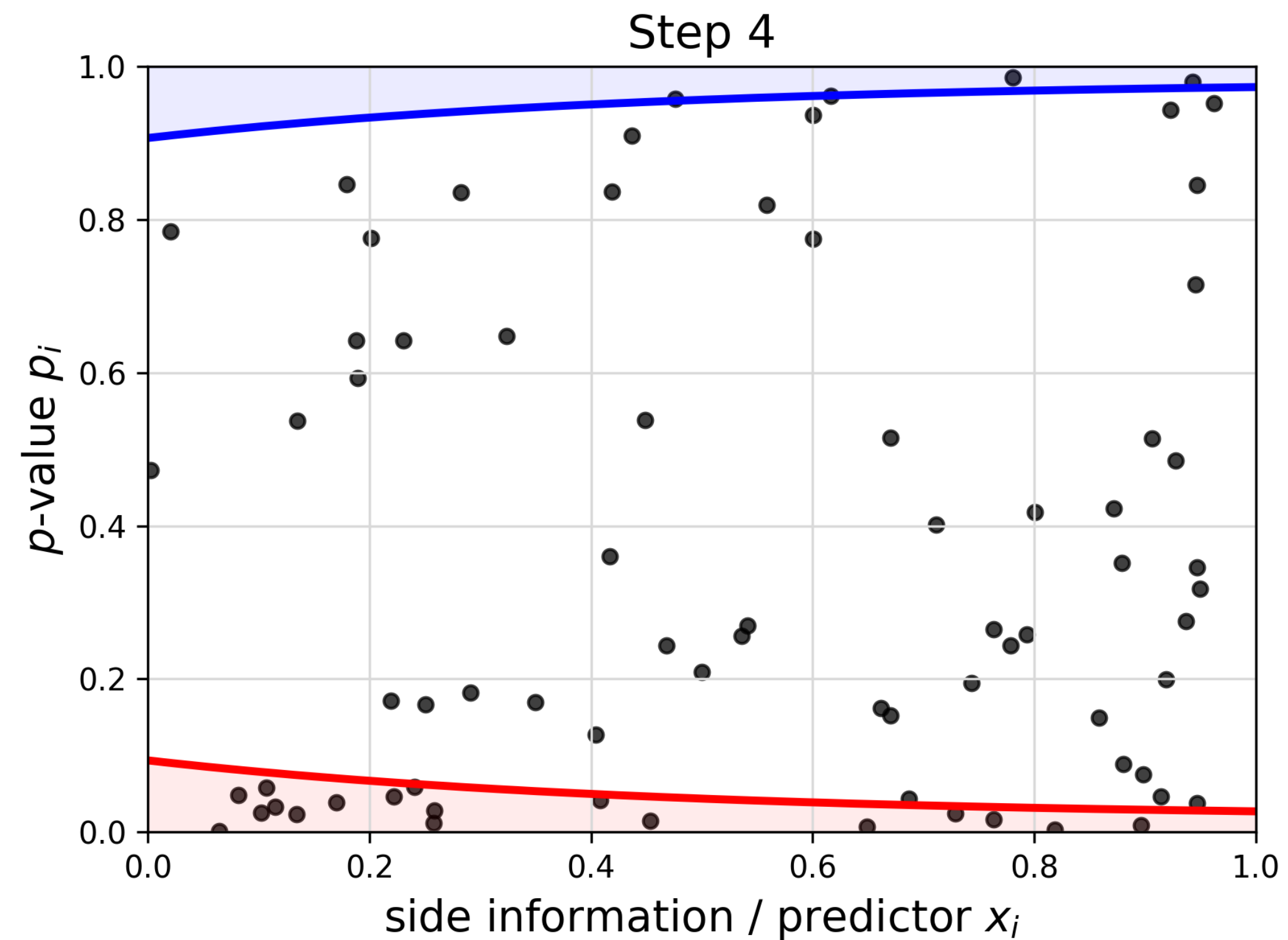
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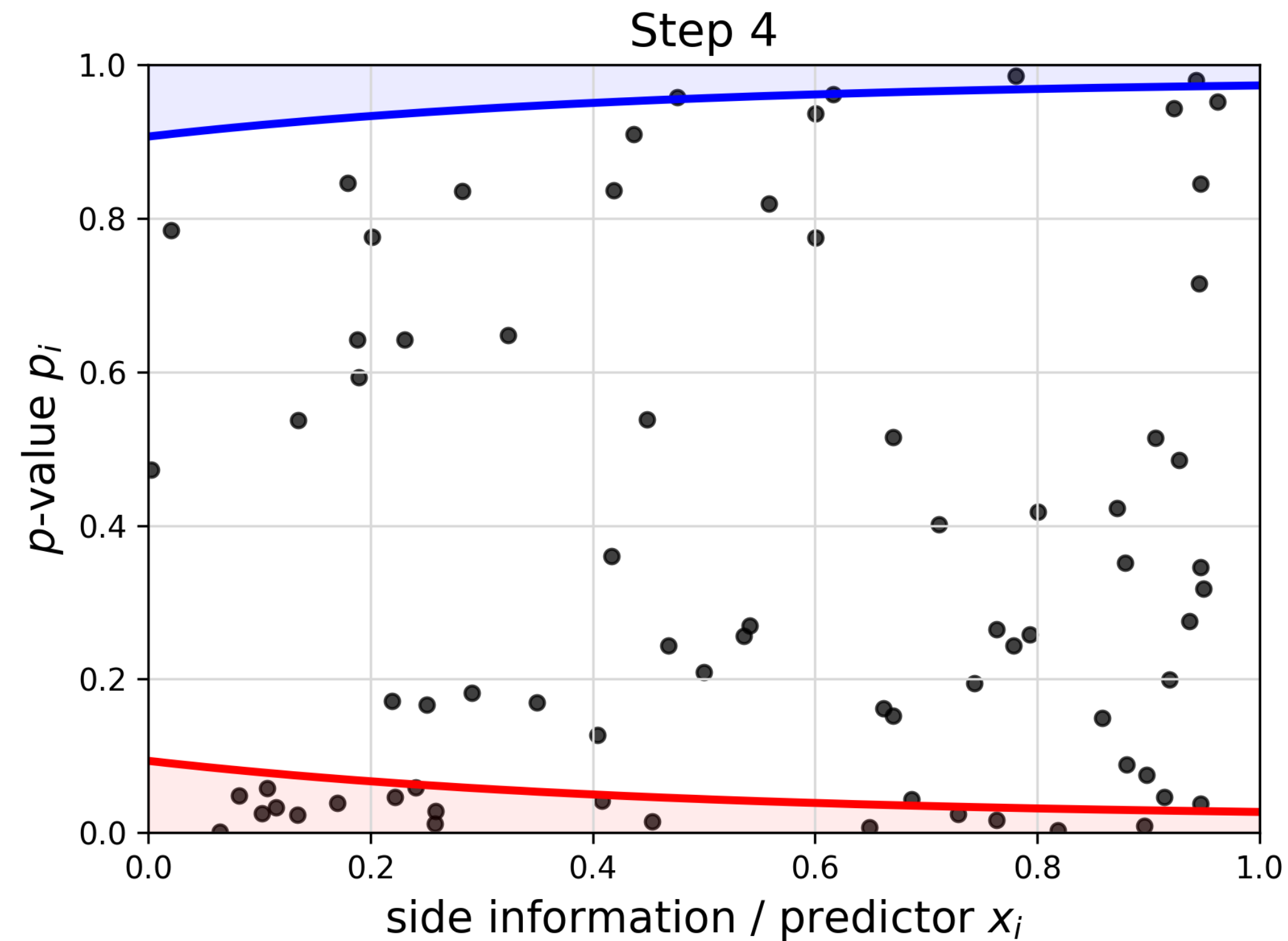


Interactive data analysis with FDR control



$$\widehat{\text{FDP}}(t) = \frac{1 + \#\{j : p_j \geq 1 - s_t(x_j)\}}{1 \vee \#\{j : p_j \leq s_t(x_j)\}}$$

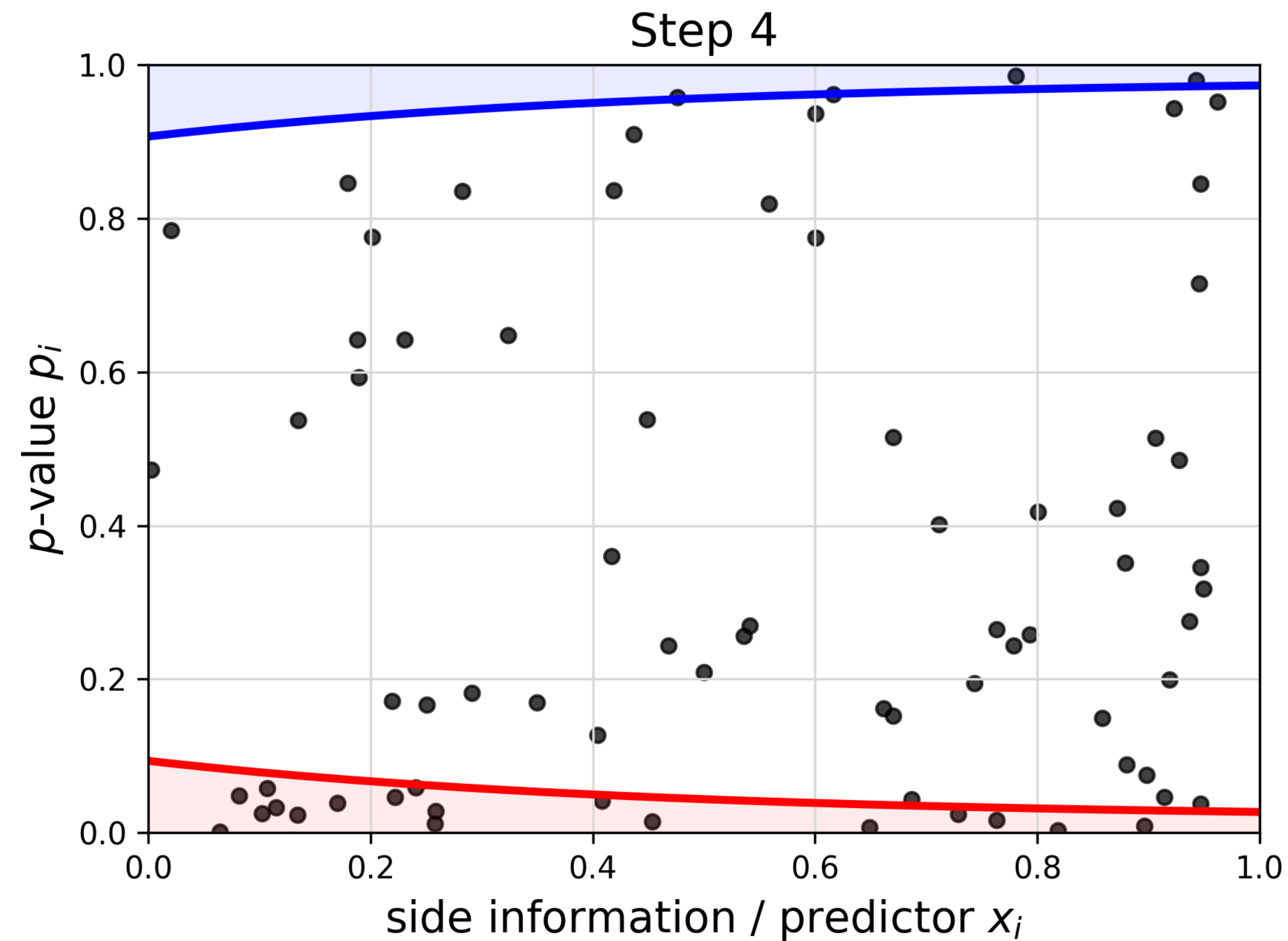
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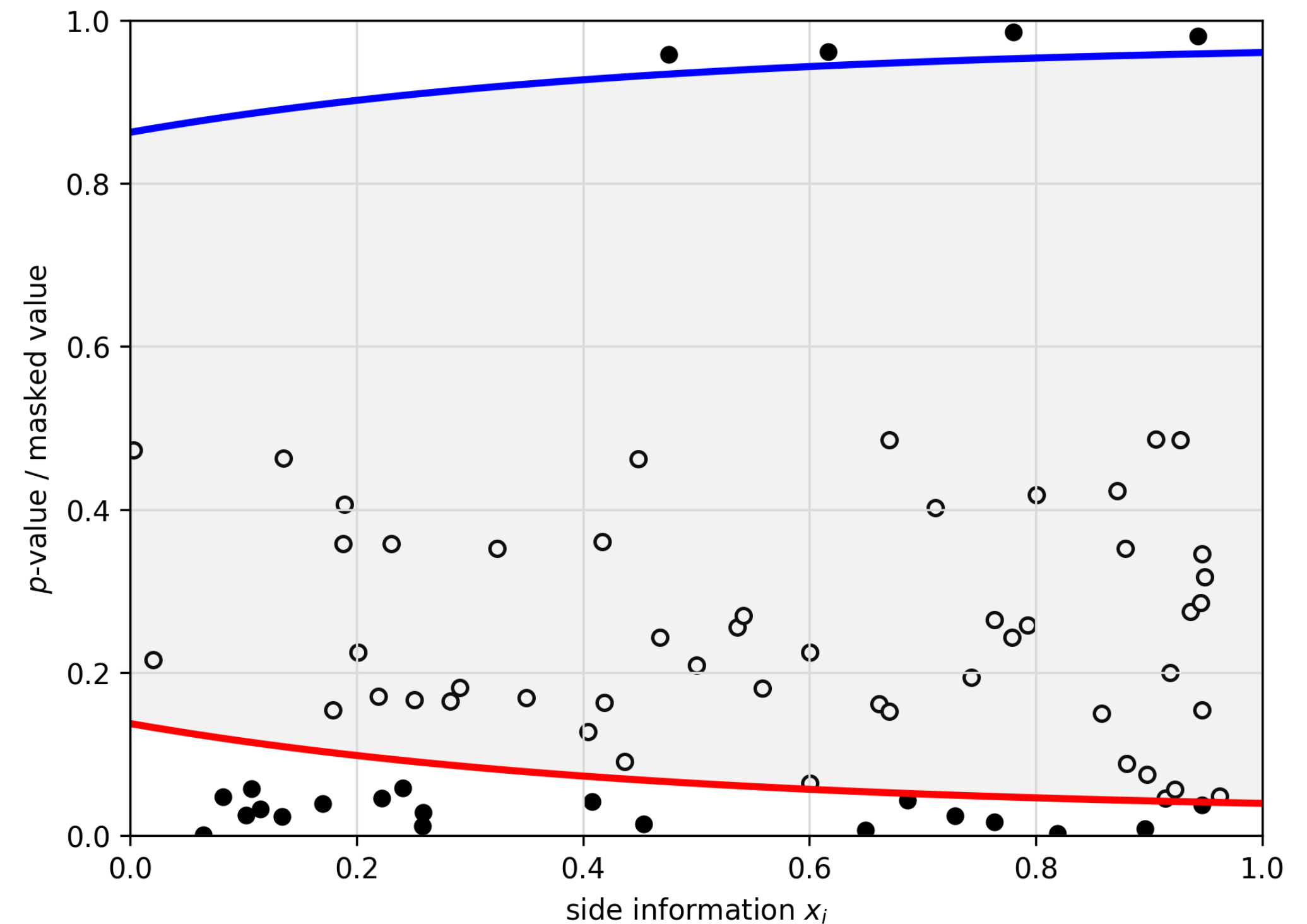
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$$\mathcal{R} = \{j : p_j \leq s_\tau(x_j)\}$$

Interactive data analysis with FDR control

Now, how to determine the form of $s_t(\cdot)$?

- ▶ Using all the data $\Rightarrow$ double dipping!
 - ▶ Instead, use a masked version of the data
- ▶ At step t , determine $s_{t+1}(\cdot)$ using
 - $\min\{p_j, 1 - p_j\}$
 - p_j for the revealed p-values at time t
 - x_j



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Example: consider a working model

$$\mathbb{P}(H_j \mid x_j) = \frac{e^{\beta^\top x_j}}{1 + e^{\beta^\top x_j}}$$

$$-\log(p_j) \mid H_j, x_j \sim \begin{cases} \text{Exp}(1) & H_j = 0 \\ \text{Exp}(\gamma^\top x_j) & H_j = 1 \end{cases}$$

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- ▶ Use the available information to estimate β, γ
- ▶ Set $s_{t+1}(\cdot)$ to be the minimum of the estimated lfdr curve and $s_t(\cdot)$

Guarantees of AdaPT

Lei and Fithian '18

When $p_1, \dots, p_m \mid x_1, \dots, x_m$ are independent, AdaPT applied at level α guarantees $\text{FDR} \leq \alpha$

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- ▶ Accommodates flexible modeling of structures
- ▶ Make decisions on the fly
- ▶ Similar to knockoffs, can be conservative when the number of signals is low

Lecture 5: Post-selection inference

Confidence interval

- ▶ For a fixed parameter θ
- ▶ A 95 % confidence interval for θ is a function of the data $C(X)$ satisfying

$$\mathbb{P}_{\theta}(\theta \in C(X)) \geq 95 \%$$

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$$\text{e.g., } \left[\bar{X} - z_{0.025} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{0.025} \frac{\sigma}{\sqrt{n}} \right]$$

Confidence interval for selected parameters

Soric's warning on reporting CI after selection

- ▶ In a large number of 95 % CIs for a fixed parameter θ
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Thought experiment

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- ▶ Observe $Z_j \sim N(\theta_j, 1)$
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Coverage is 0!

False coverage rate (FCR)⁹

“Coverage averaged over the selected”

$$\text{FCR} = \mathbb{E} \left[\frac{\#\{j \in \mathcal{R}, \theta_j \notin C_j\}}{\max\{|R|, 1\}} \right]$$

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$$\text{FCR} = \mathbb{E} \left[\frac{\sum_{j=1}^m 1\{\theta_j \notin C_j^\alpha\}}{m} \right] = \frac{1}{m} \sum_{j=1}^m \mathbb{P}(\theta_j \notin C_j^\alpha) \leq \alpha$$

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Can be quite conservative

FCR-adjusted CIs

- ▶ Observe $Z_j \stackrel{\text{ind}}{\sim} N(\theta_j, 1)$
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- ▶ Selection rule: BH

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A thought experiment: suppose $z_{\alpha|\mathcal{R}|/m} = 3.5$, $Z_j = 15 \leadsto$ FCR-adjusted CI $[11.5, 18.5]$

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But...

- ▶ If Z_j is already optimistic, why still inflating it after selection?
- ▶ The inflation is symmetric while the selection procedure is biased in one direction

Selection-conditional CIs

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Controls $\text{FCR} \leq \alpha$

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truncated Gaussian distribution

Selection-condition CI: [2.5% quantile, 97.5% quantile]

Summary

- ▶ The issue of multiplicity
- ▶ Global null testing
- ▶ Multiple testing and different error metrics
- ▶ Adaptive/interactive inference
- ▶ Post-selection inference

Thank you!

Email: zren@wharton.upenn.edu